

Actuation systems

# Introduction

- Actuation systems are the elements of control systems which are responsible for transforming the output of a microprocessor or control system into a controlling action on a machine or device.

Example : an electrical output from the controller which has to be transformed into a linear motion to move a load.

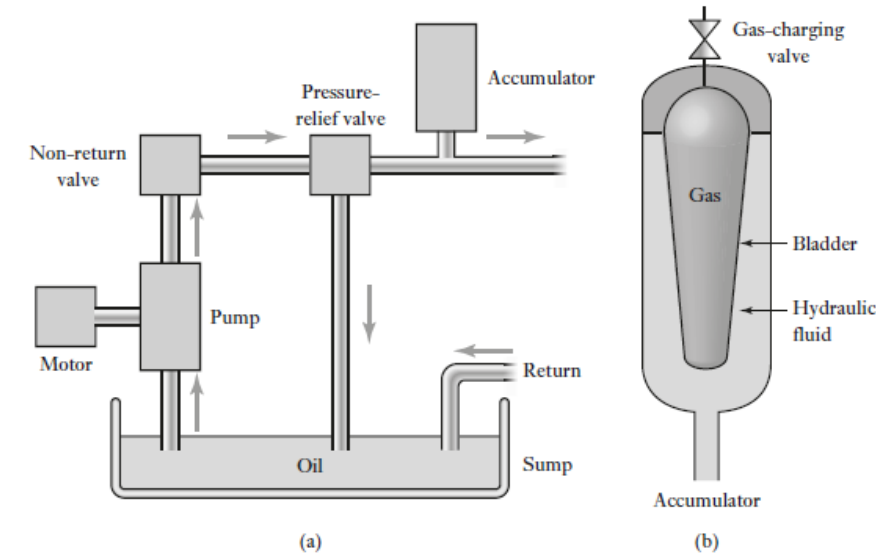
Example : an electrical output from the controller has to be transformed into an action which controls the amount of liquid passing along a pipe.

- ❖ Pneumatic and hydraulic actuation systems
- ❖ mechanical actuator systems
- ❖ electrical actuation systems

# Hydraulic actuation systems

- Hydraulic system uses pressurized oil is provided by a pump driven by an electric motor

Figure 7.1 (a) Hydraulic power supply, (b) accumulator.



- The pump, pumps oil from a sump through a non-return valve and an accumulator to the system, from which it returns to the sump
- A *pressure-relief valve releases the pressure if it rises above a safe level*, the *non-return valve is to prevent the oil being back driven to the pump* and the *accumulator is to smooth out any short-term fluctuations in the output oil pressure*.
- Pressurized gas supplied to the bladder and the chamber containing the hydraulic fluid
- If the oil pressure rises then the bladder contracts, increases the volume the oil can occupy and so reduces the pressure.
- If the oil pressure falls, the bladder expands to reduce the volume occupied by the oil and so increases its pressure.

# HYDRAULIC PUMPS

- Pumps are used to supply the high pressure that the mechatronic system requires.

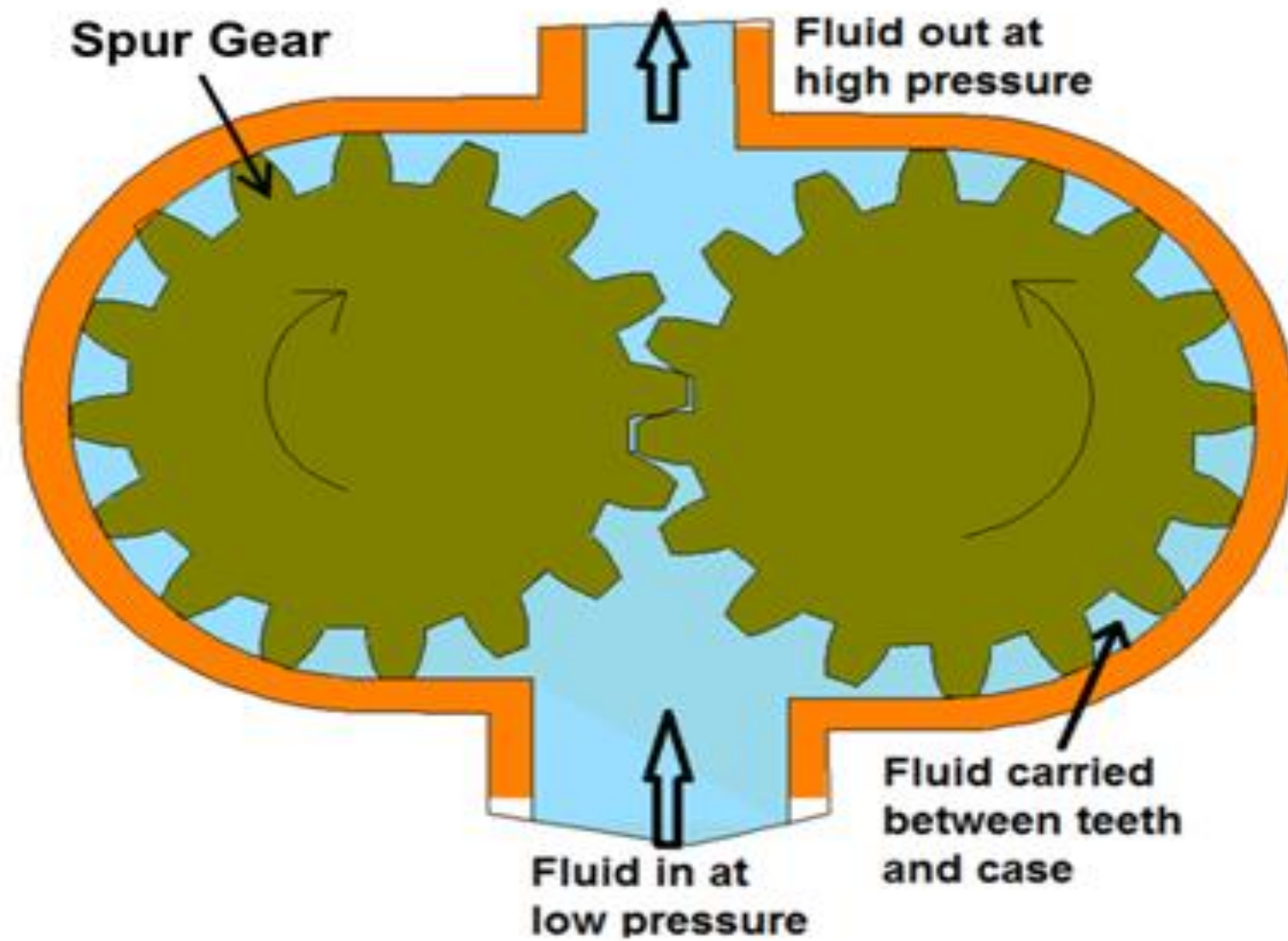
Three types are most commonly used:

- Gear pump;
- Vane pump; and
- Piston pump.

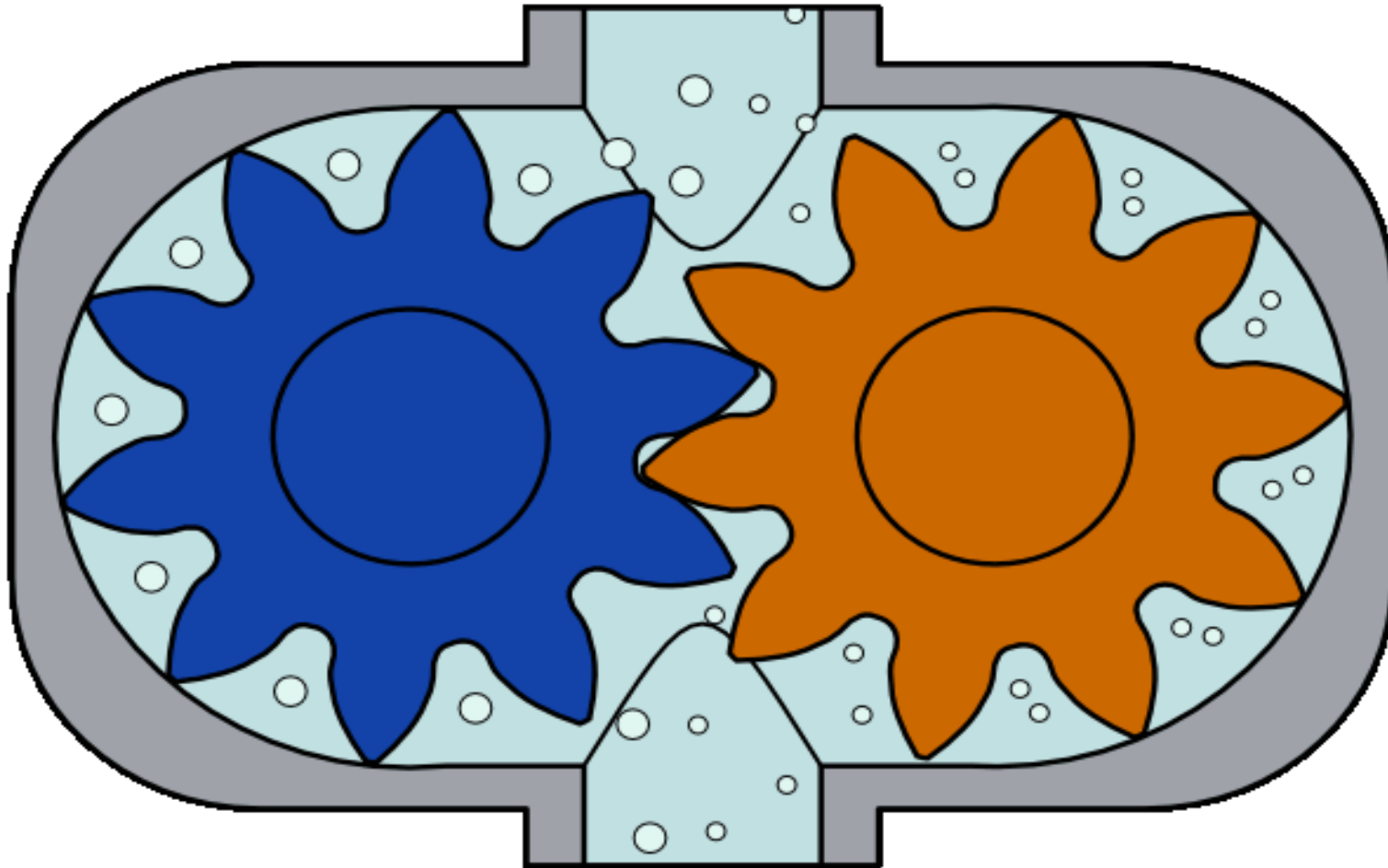
# Gear Pump:

- Two close-meshing gear wheels which rotate in opposite directions.
- One of the gears is coupled with a prime mover and is called as driving gear and another is called as driven gear.
- When the gears rotate, volume of the chamber expands leading to pressure drop below atmospheric value.
- Therefore the vacuum is created and the fluid is pushed into the void due to atmospheric pressure.
- The fluid is trapped between housing and rotating teeth of the gears.
- The leakage occurs between the teeth and the casing and between the interlocking teeth limits the efficiency

# Gear Pump:



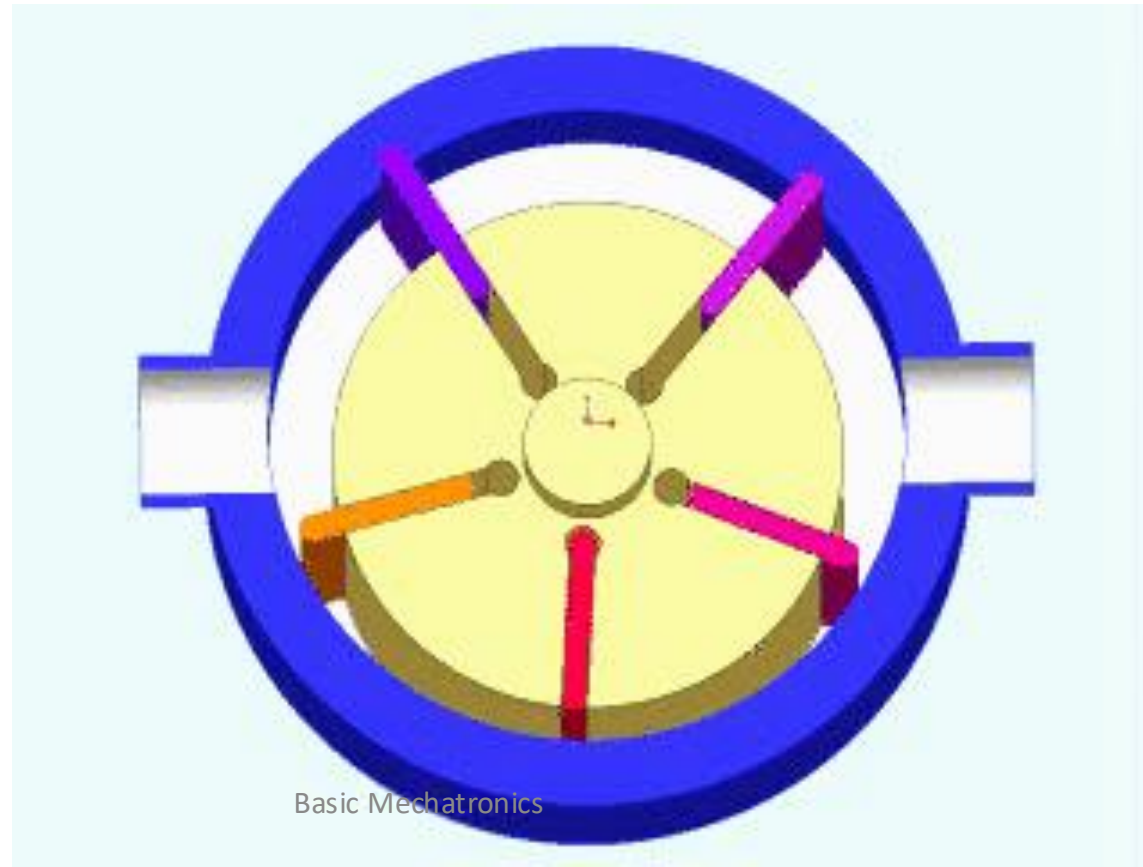
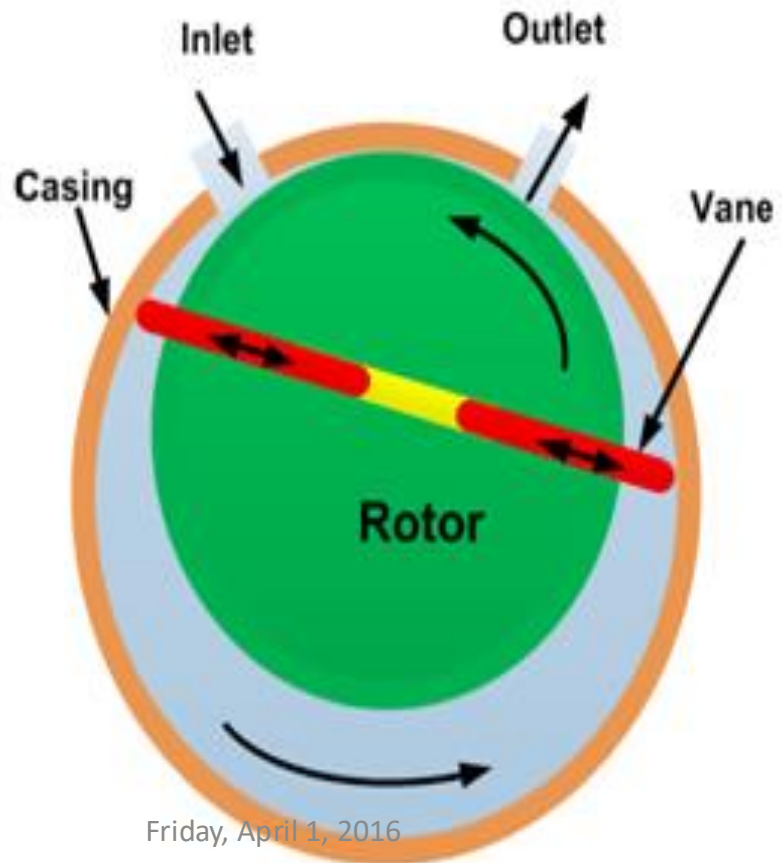
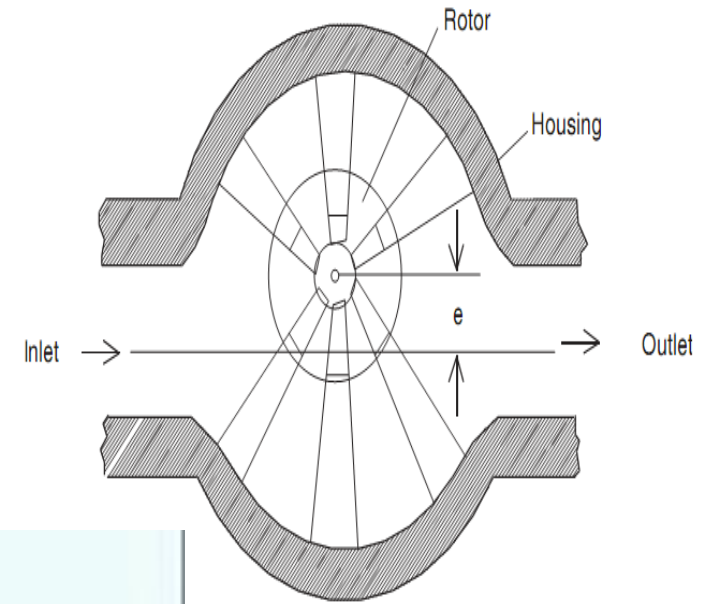
# Gear Pump:



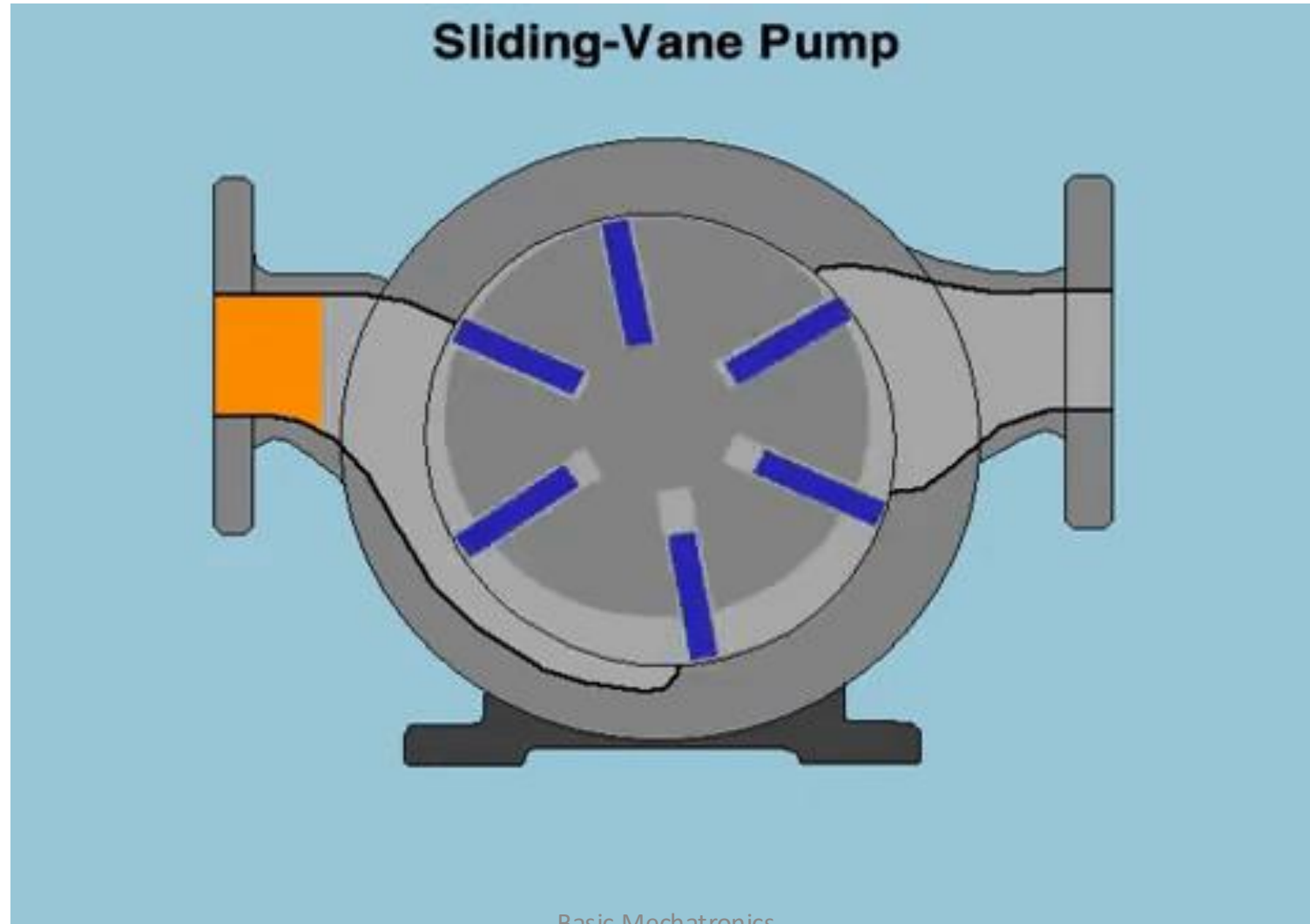
# Vane Pump:

- Vane pumps generate a pumping action by tracking of vanes along the casing wall.
- The vane pump has *spring-loaded sliding vanes slotted in a driven rotor*
- The vane pumps generally consist of a rotor, vanes, ring and a port plate with inlet and outlet ports.
- The rotor in a vane pump is connected to the prime mover through a shaft. The vanes are located on the slotted rotor.
- *As the rotor rotates, the vanes follow the contours of the casing. This results in fluid becoming trapped between successive vanes and the casing and transported round from the inlet port to outlet port.*
- The leakage is less than with the gear pump.

# Vane Pump:

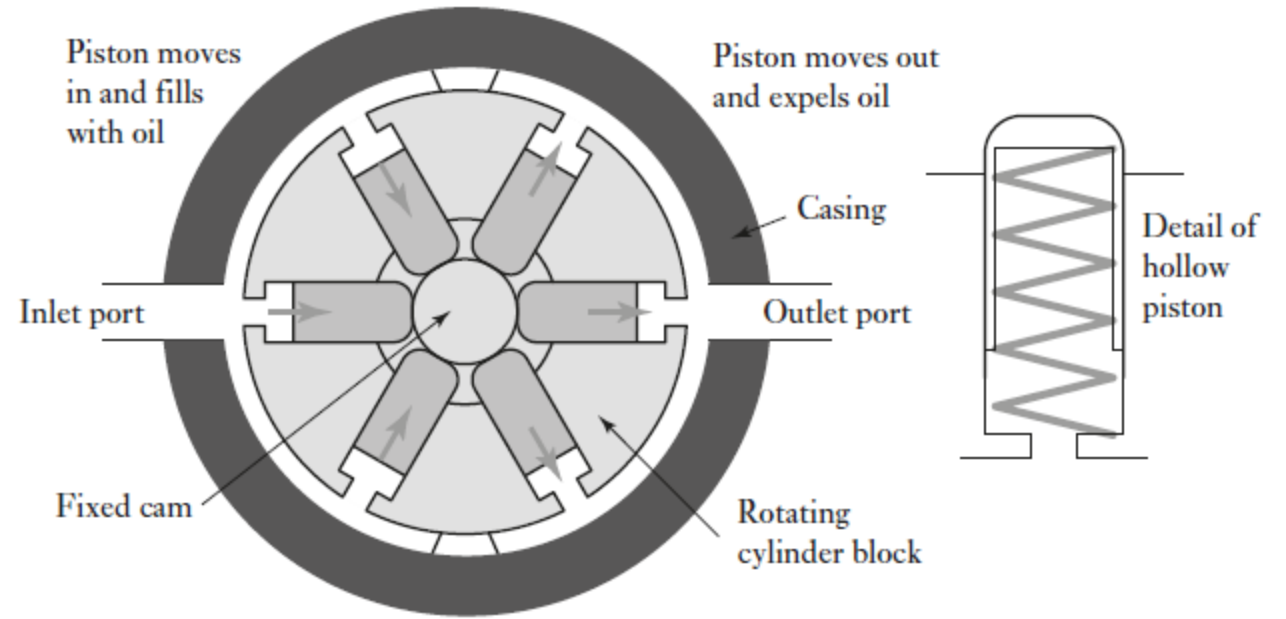


# Vane Pump:



# Piston Pump:

## Radial piston pump

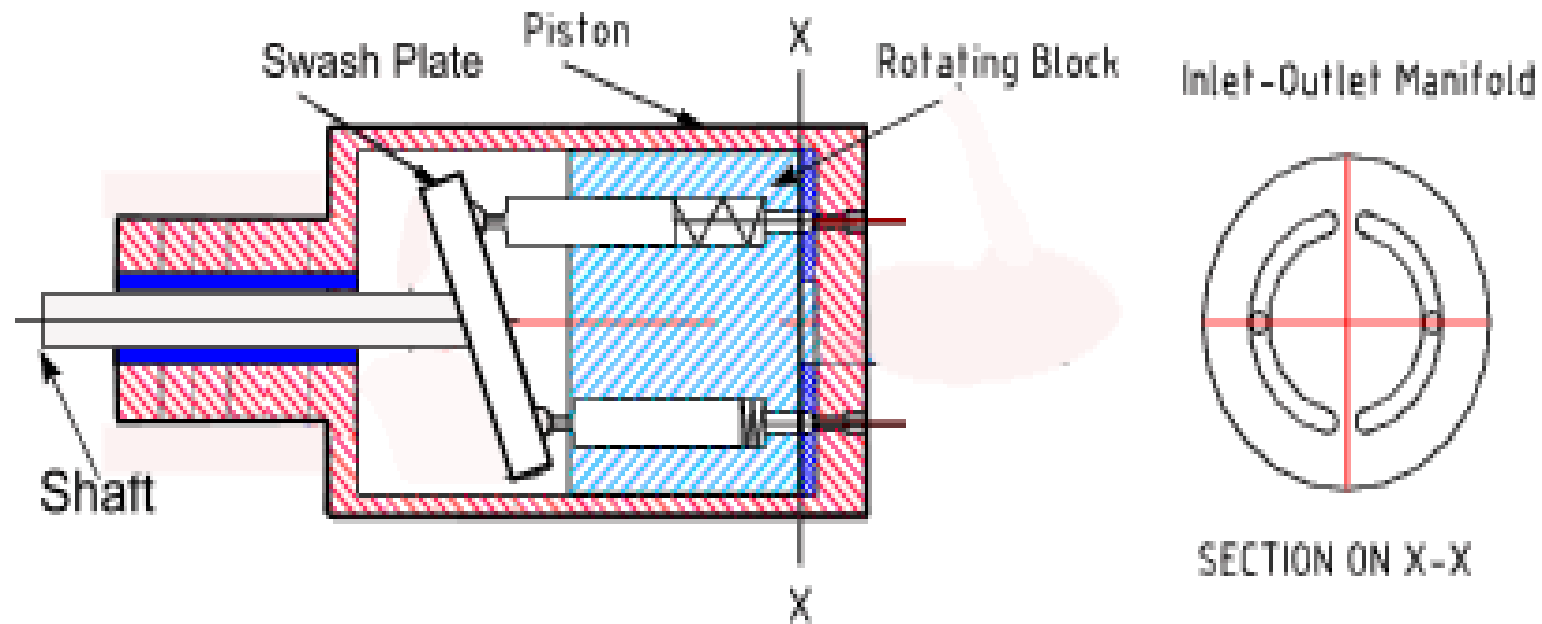
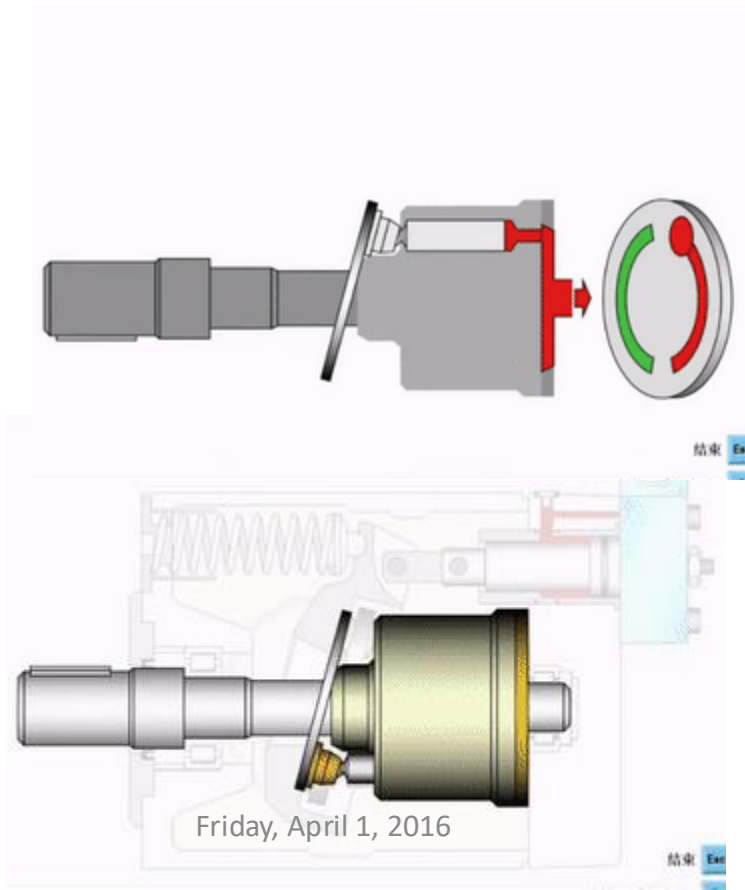
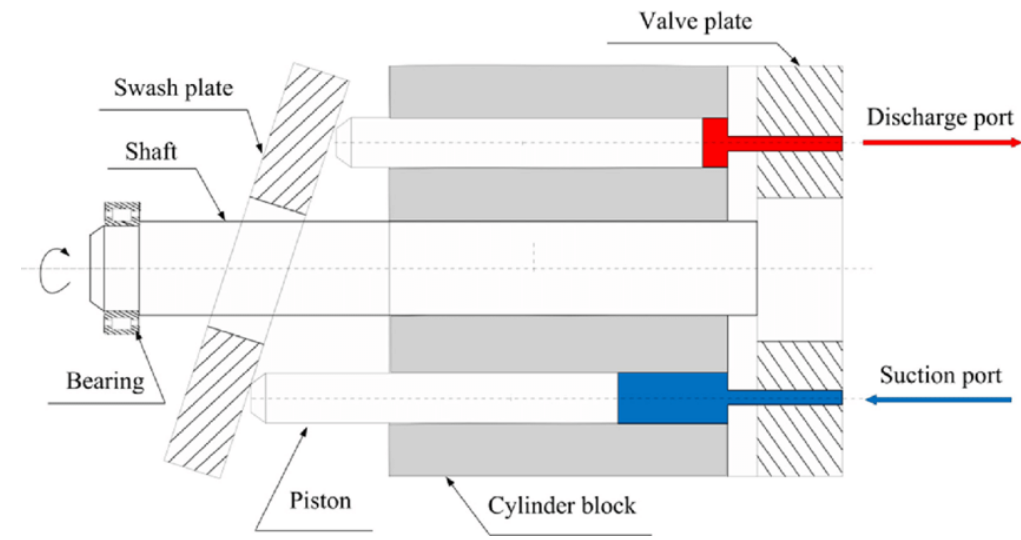


- a cylinder block rotates round the stationary cam
- Hollow pistons, with spring return, to move in and out
- The fluid is drawn through the inlet port and transported round and ejection through the discharge port

# Piston Pump

## Axial piston pump

- Pistons which move axially rather than radially



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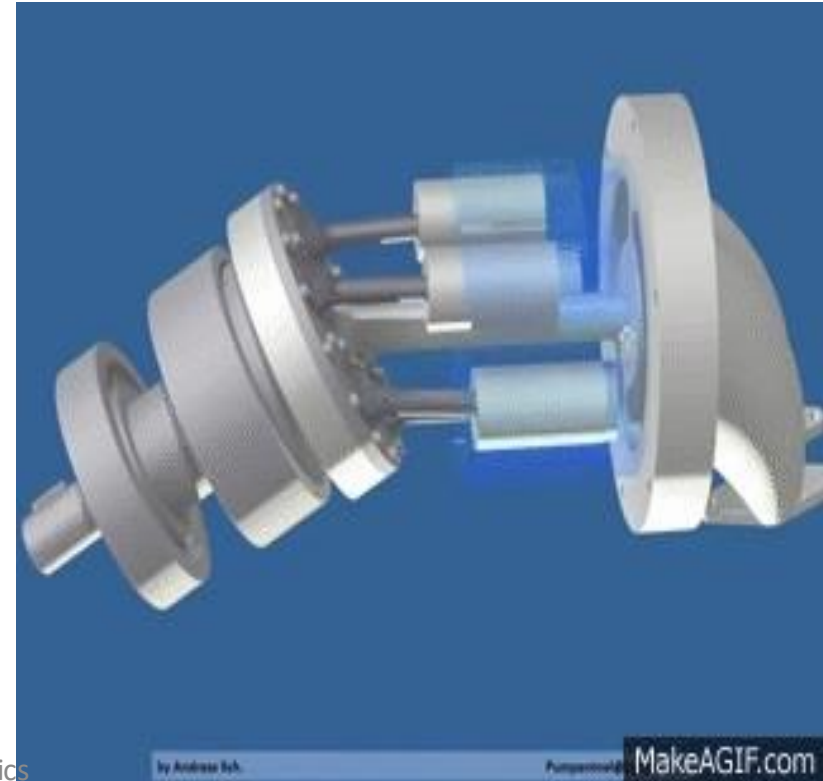
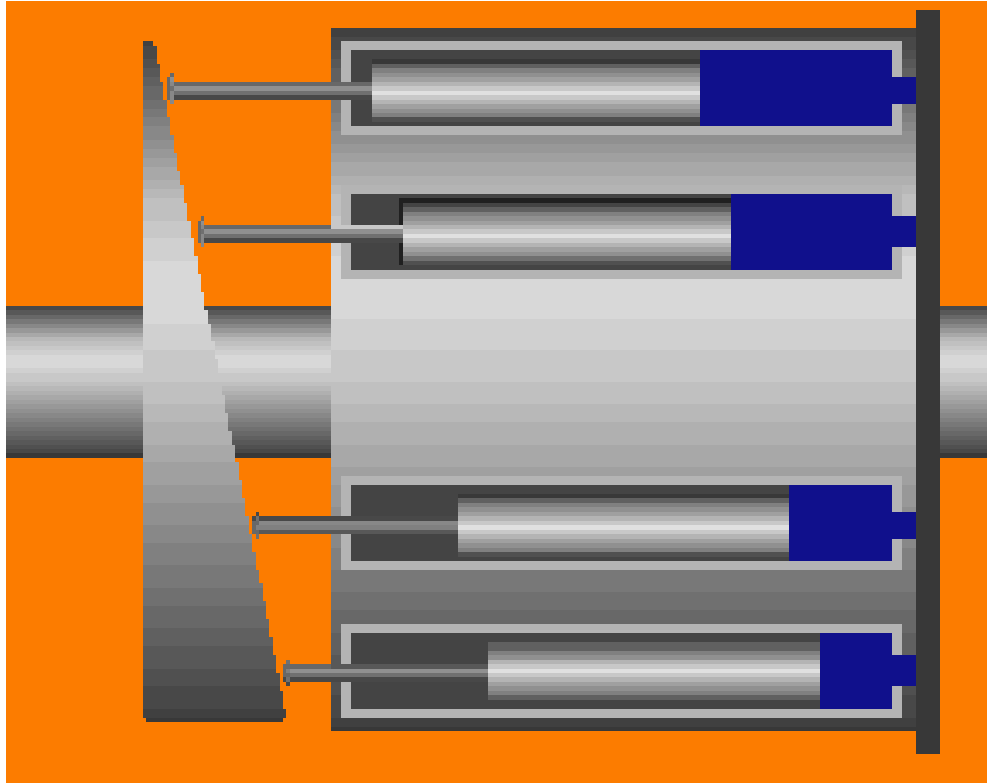
SwashPlate Motor

All components rotate except case and Manifold Plate

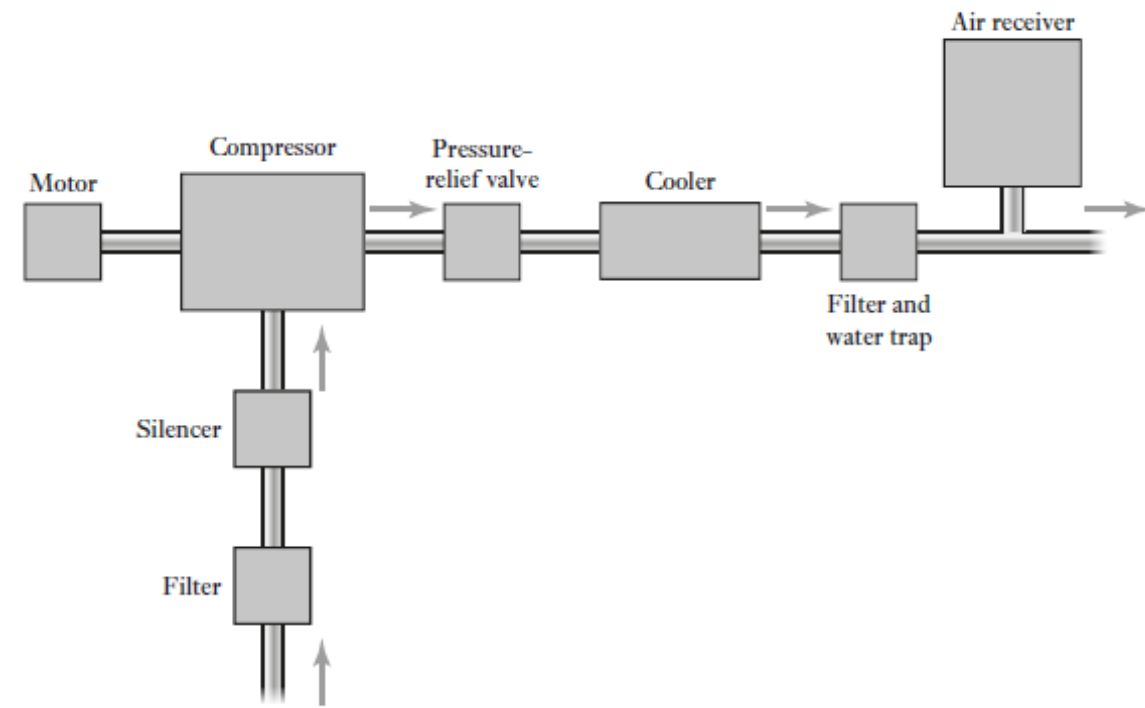
# Axial piston pump

- Axial piston pumps are positive displacement pumps which convert rotary motion of the input shaft into an axial reciprocating motion of the pistons.
- These pumps have a number of pistons (usually an odd number) in a circular array within a housing which is commonly referred to as a cylinder block, rotor or barrel.
- Pistons are arranged axially in a rotating cylinder block and move by contact with the swash plate.
- This plate is at an angle to the drive shaft
- The shaft rotates and moves the pistons so that fluid is sucked in when a piston is opposite the inlet port and expelled when it is opposite the discharge port.
- Piston pumps have a high efficiency and can be used at higher hydraulic pressures than gear or vane pumps.

# Axial Piston Pump:



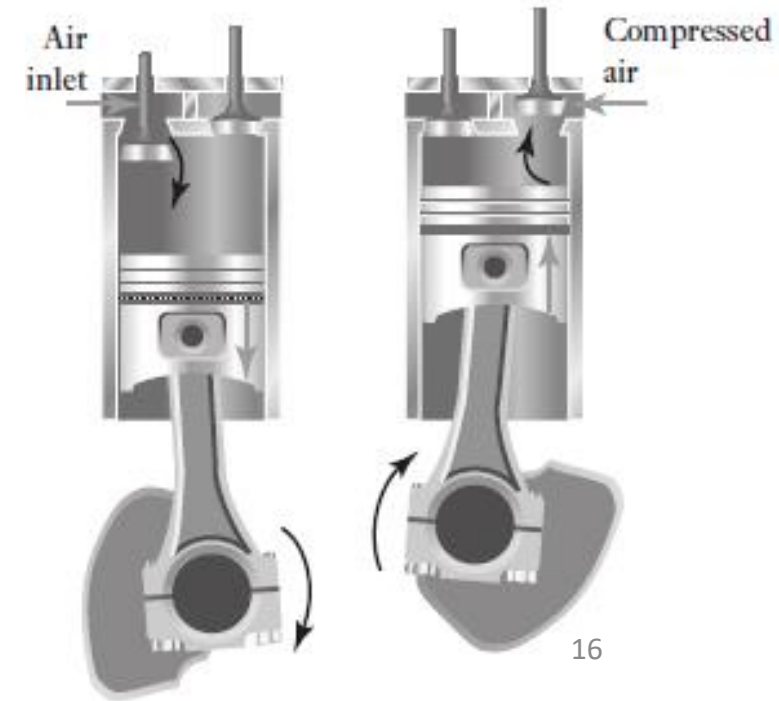
# Pneumatic actuation systems



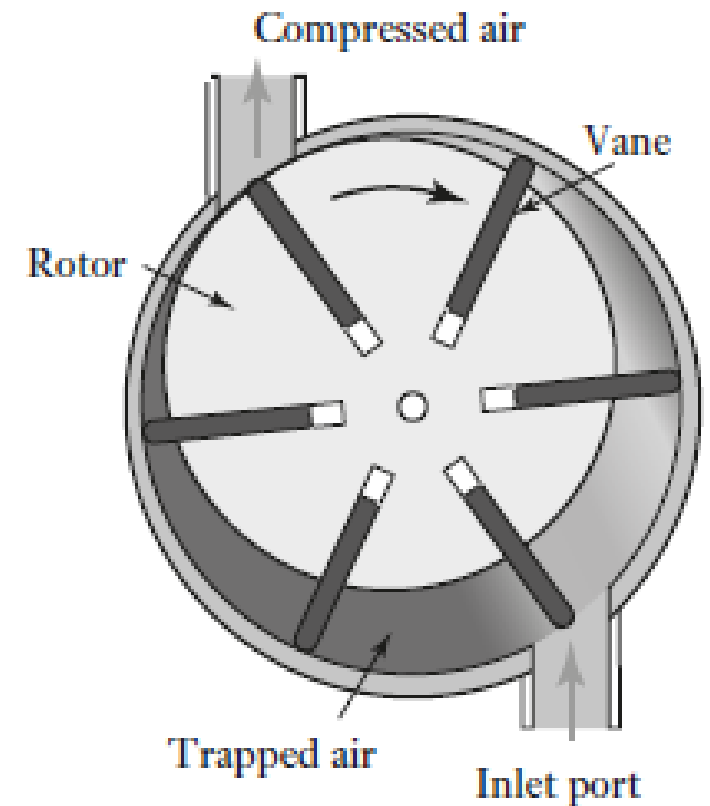
- Working with a pneumatic power supply
- An electric motor drives an air compressor
- The air inlet to the compressor is likely to be filtered and via a silencer to reduce the noise level.
- A pressure-relief valve provides protection against the pressure in the system rising above a safe level.
- The air compressor increases the temperature of the air, there is likely to be a cooling system
- To remove contamination and water from the air a filter with a water trap.
- An air receiver smooth-out any short-term pressure fluctuations.

# Single-acting single-stage reciprocating compressor

- The air intake stroke, the descending piston causes air to be sucked into the chamber through the spring-loaded inlet valve
- The trapped air forces the inlet valve to close and so air becomes compressed.
- When the air pressure has risen sufficiently, the spring-loaded outlet valve opens and the trapped air flows into the compressed-air system.
- Pressure up to 140 bar, air flow deliveries range from 0.02 m<sup>3</sup>/min to 600 m<sup>3</sup>/min

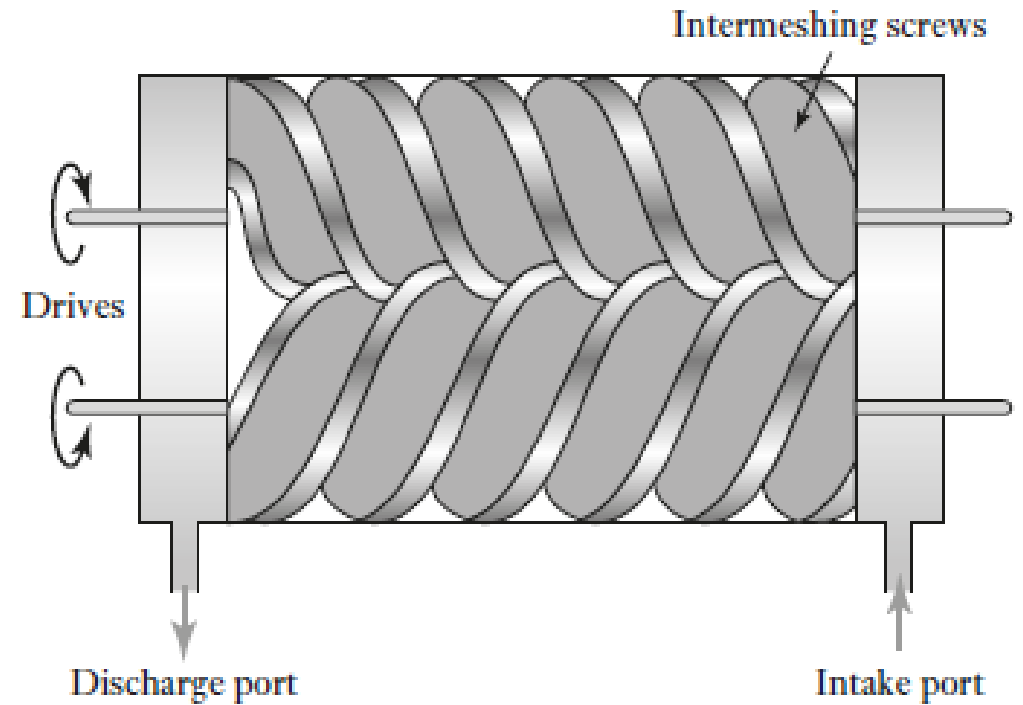
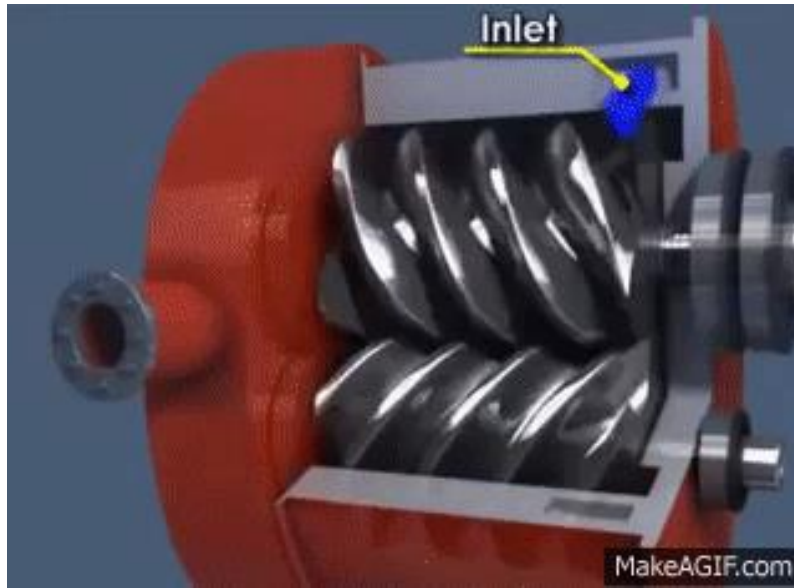


# Rotary vane compressor



- Rotor mounted eccentrically in a cylindrical chamber
- The rotor has blades, the vanes, which are free to slide in radial slots with rotation causing the vanes to be driven outwards against the walls of the cylinder
- As the rotor rotates, air is trapped in pockets formed by the vanes and as the rotor rotates so the pockets become smaller and the air is compressed.
- Compressed air are thus discharged from the discharge port.
- Pressures up to about 800 kPa with flow rates of the order of 0.3 m<sup>3</sup>/min to 30 m<sup>3</sup>/min

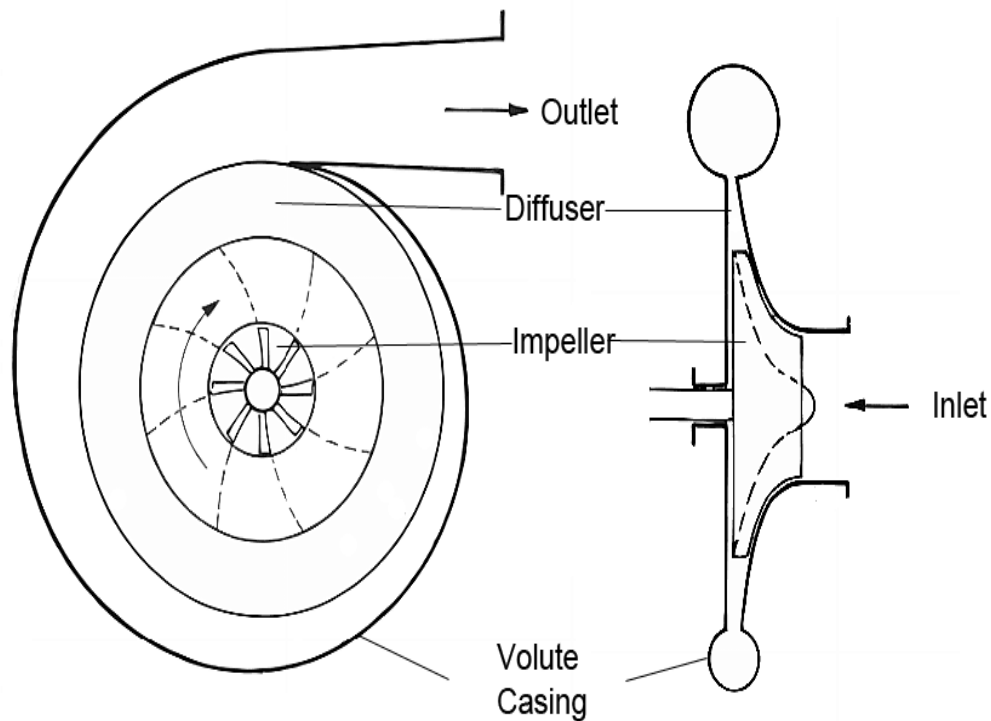
# Rotary screw compressor



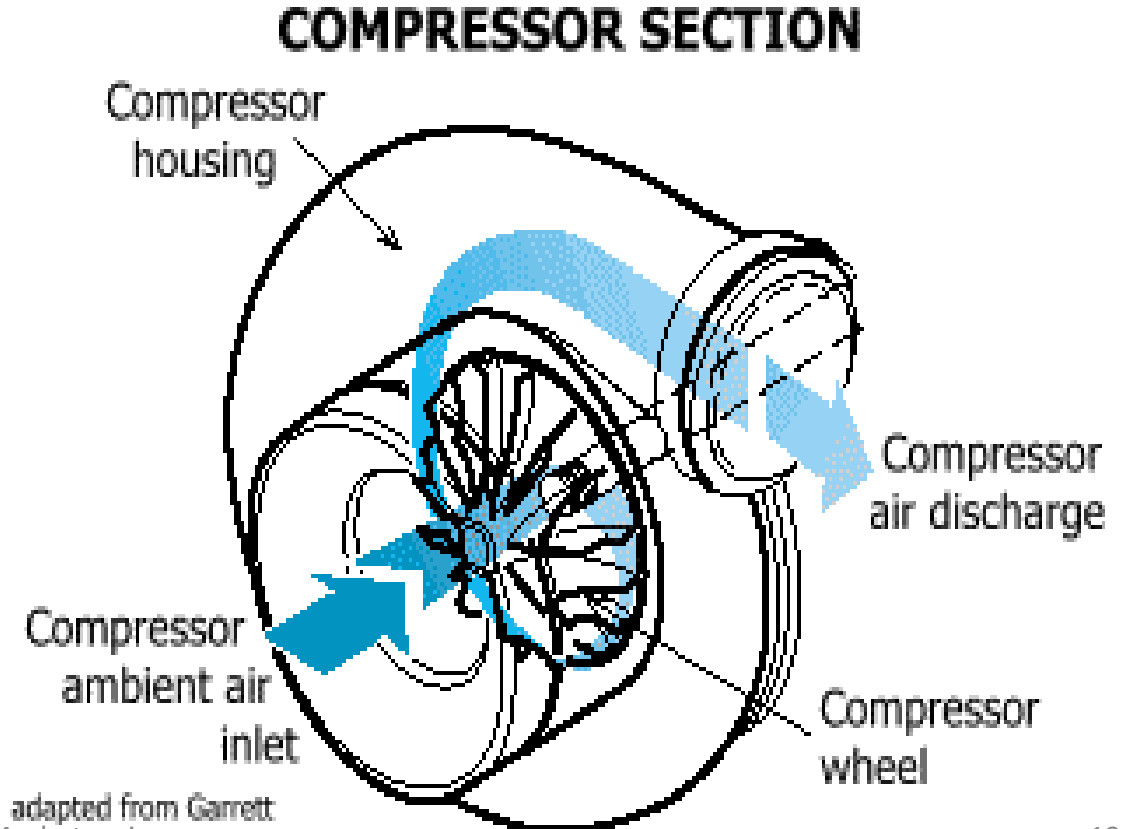
- This has two intermeshing rotary screws which rotate in opposite directions.
- As the screws rotate, air is drawn into the casing through the inlet port and into the space between the screws
- This trapped air is moved along the length of the screws and compressed as the space becomes progressively smaller
- Emerging from the discharge port.
- Pressures up to about 1000 kPa with flow rates of between 1.4 m<sup>3</sup>/min and 60 m<sup>3</sup>/min

# Centrifugal compressors

- In centrifugal compressors, air enters the eye or centre of the impeller
- Centrifugal effect throws the air into the volute where it goes to the diffuser

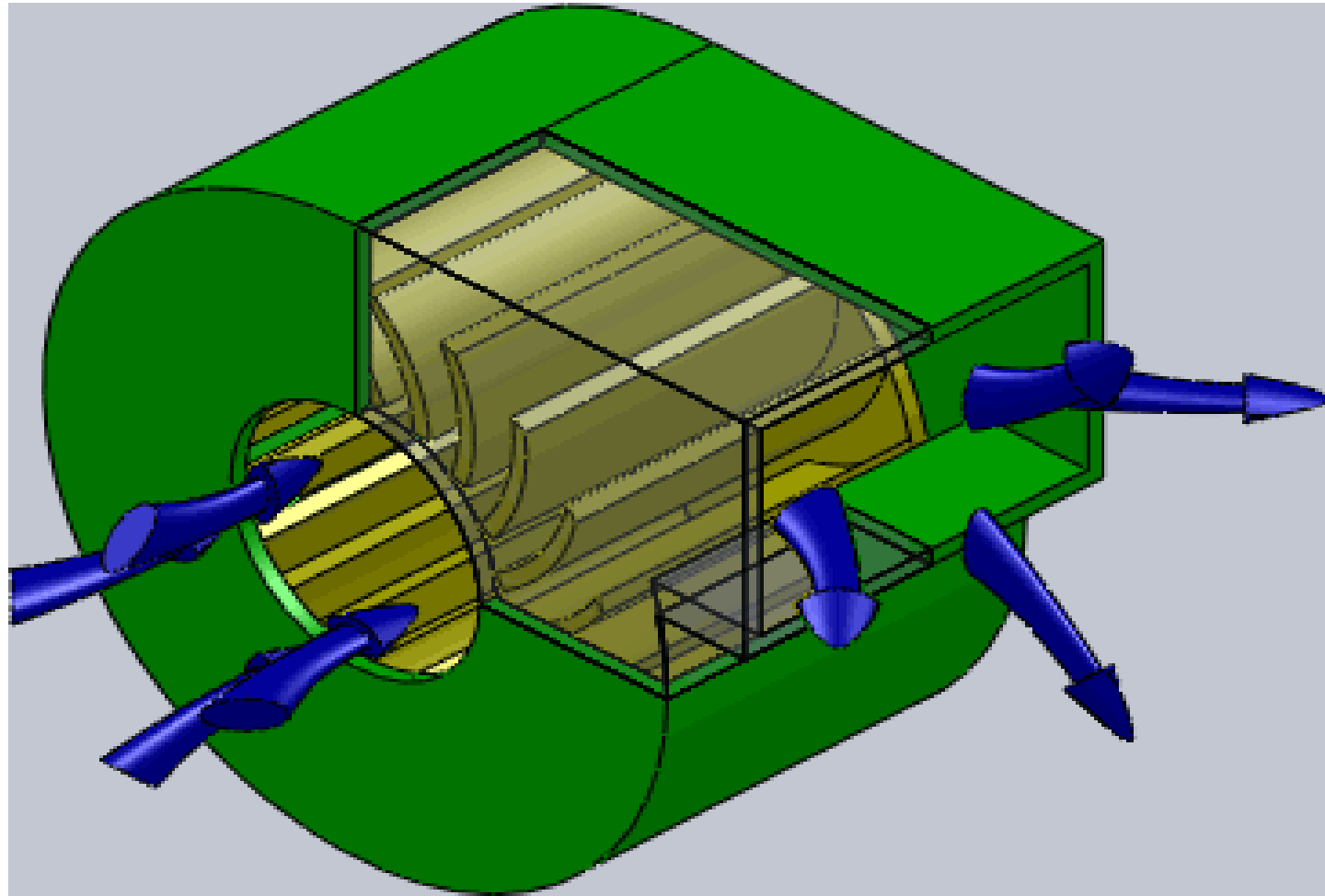


Centrifugal compressor schematic diagram



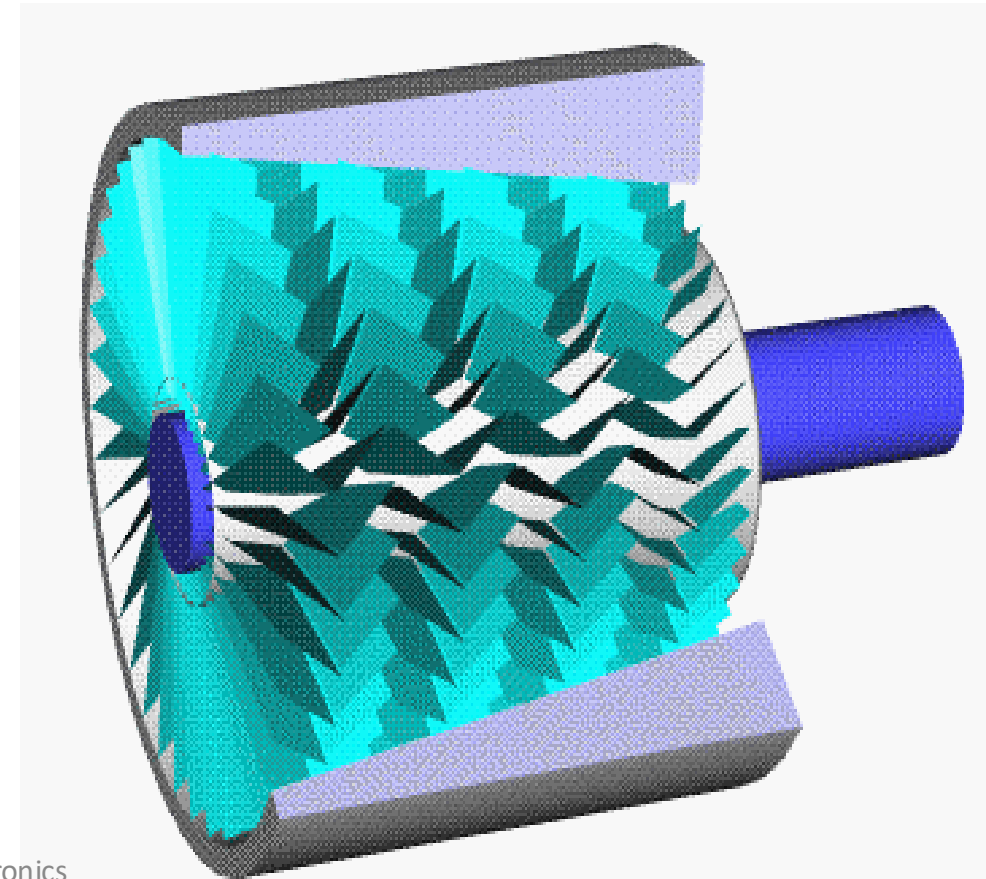
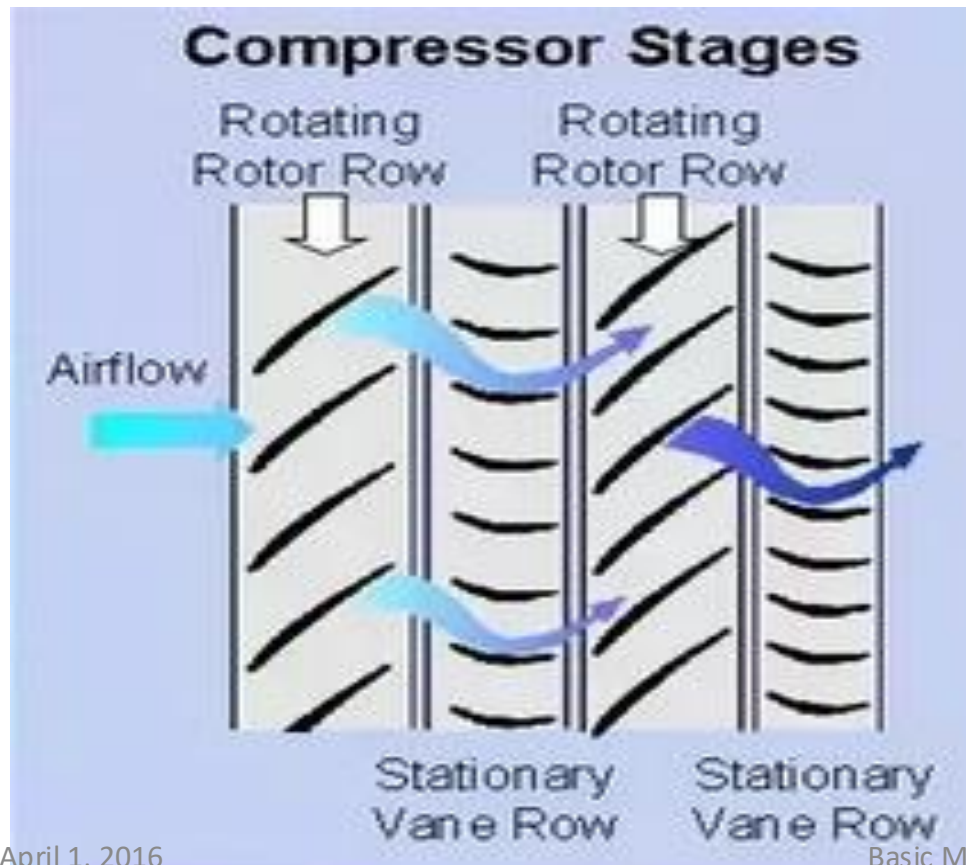
adapted from Garrett  
Basic Mechatronics

# Centrifugal compressors



# Axial compressor

- In axial compressors, the annular inlet area is much greater than that of the eye of the centrifugal type. Hence it delivers more flow.
- The blades are attached to the tapered rotor.



# Linear Hydraulic Actuators - Cylinders

- Hydraulic linear actuators, as their name implies, provide motion in a straight line.
- They are usually referred to as cylinders, rams and jacks.
- *The function of hydraulic cylinder is to convert hydraulic power into linear mechanical force or motion.*
- Hydraulic cylinders extend and retract a piston rod to provide a push or pull force to drive the external load along a straight-line path.

# Types of Linear Hydraulic Actuators:

Hydraulic cylinders are of the following types:

- Single-acting cylinders
- Double-acting cylinders
- Telescopic cylinders.
- Tandem cylinders.

# Single Acting Cylinders:

- A single-acting cylinder is simplest in design.
- *These cylinders use hydraulic pressure to extend the piston rod in one direction and rely on an external force, such as a spring, to retract it.*

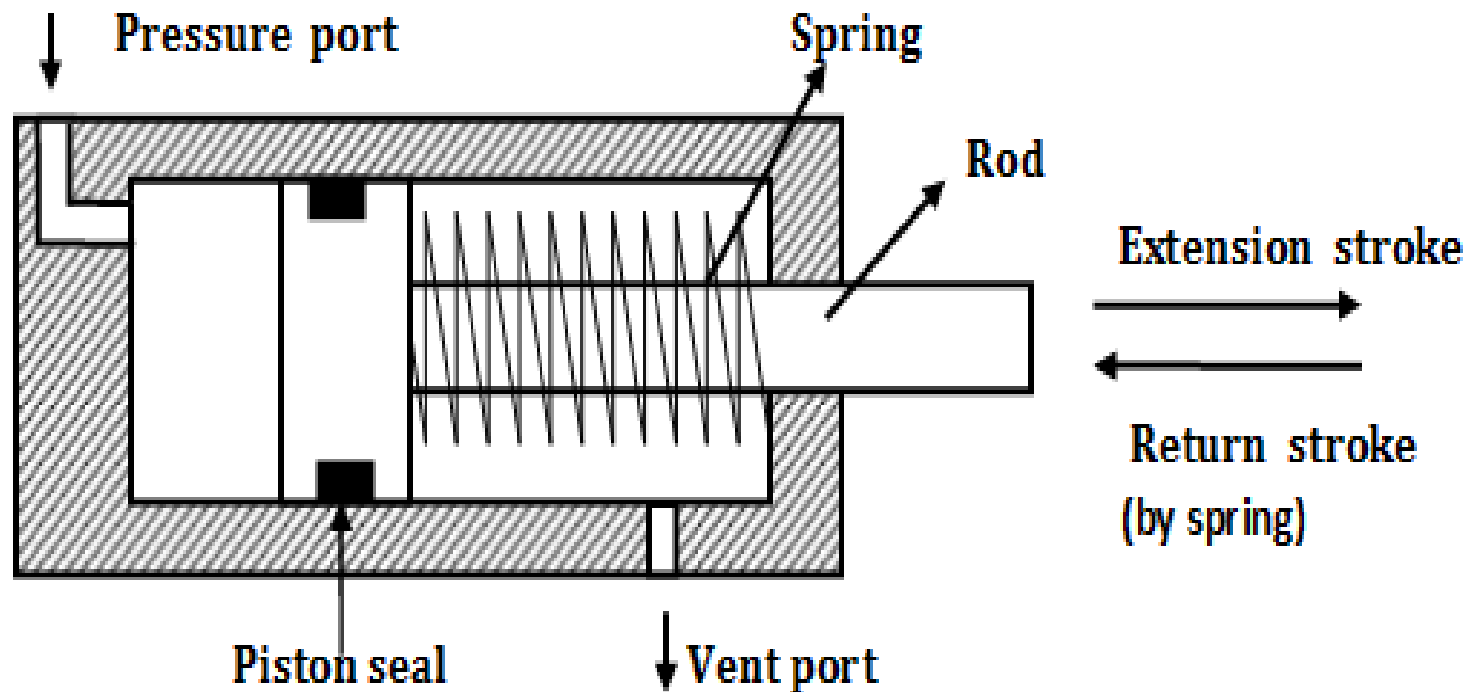
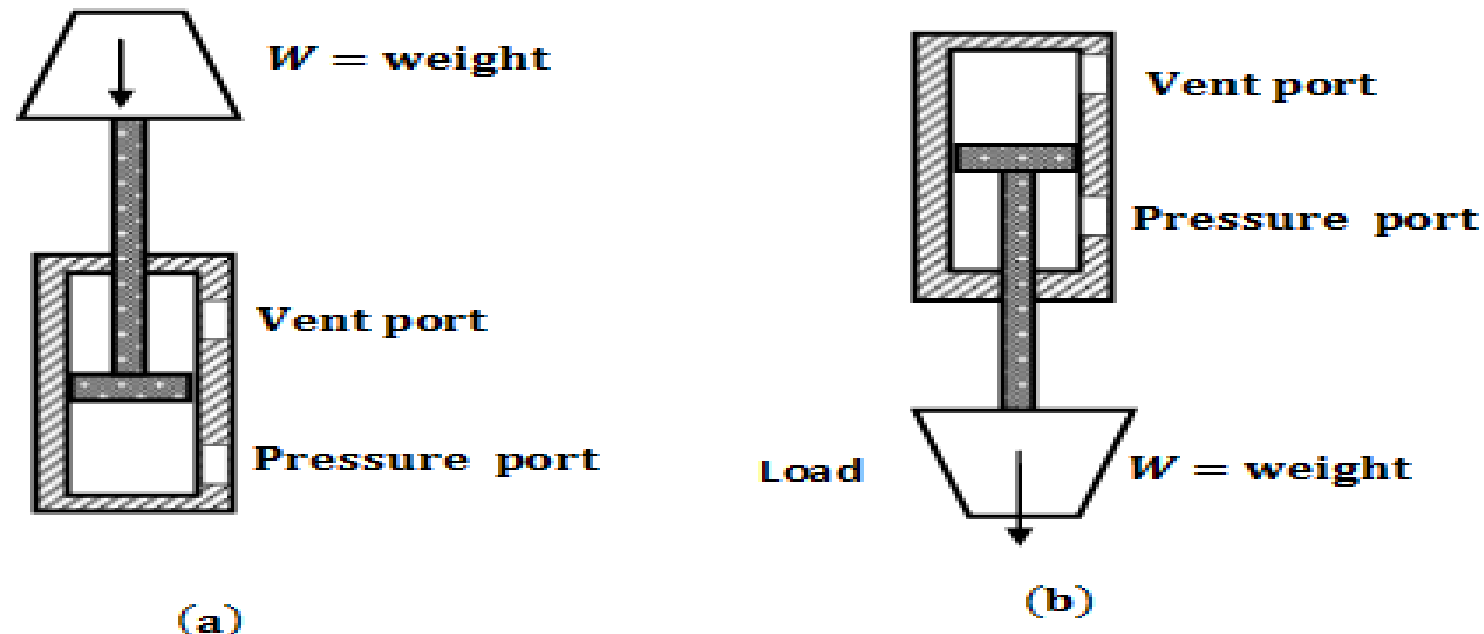


Figure 1.1 Single-acting cylinders  
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According to the type of return, single-acting cylinders are classified as follows:

- Gravity-return single-acting cylinder.
- Spring-return single-acting cylinder.

**Gravity-Return Single-Acting Cylinder**



**Figure 1.2** Gravity-return single-acting cylinder: (a) Push type; (b) pull type

- Spring Return Single Acting Cylinder:

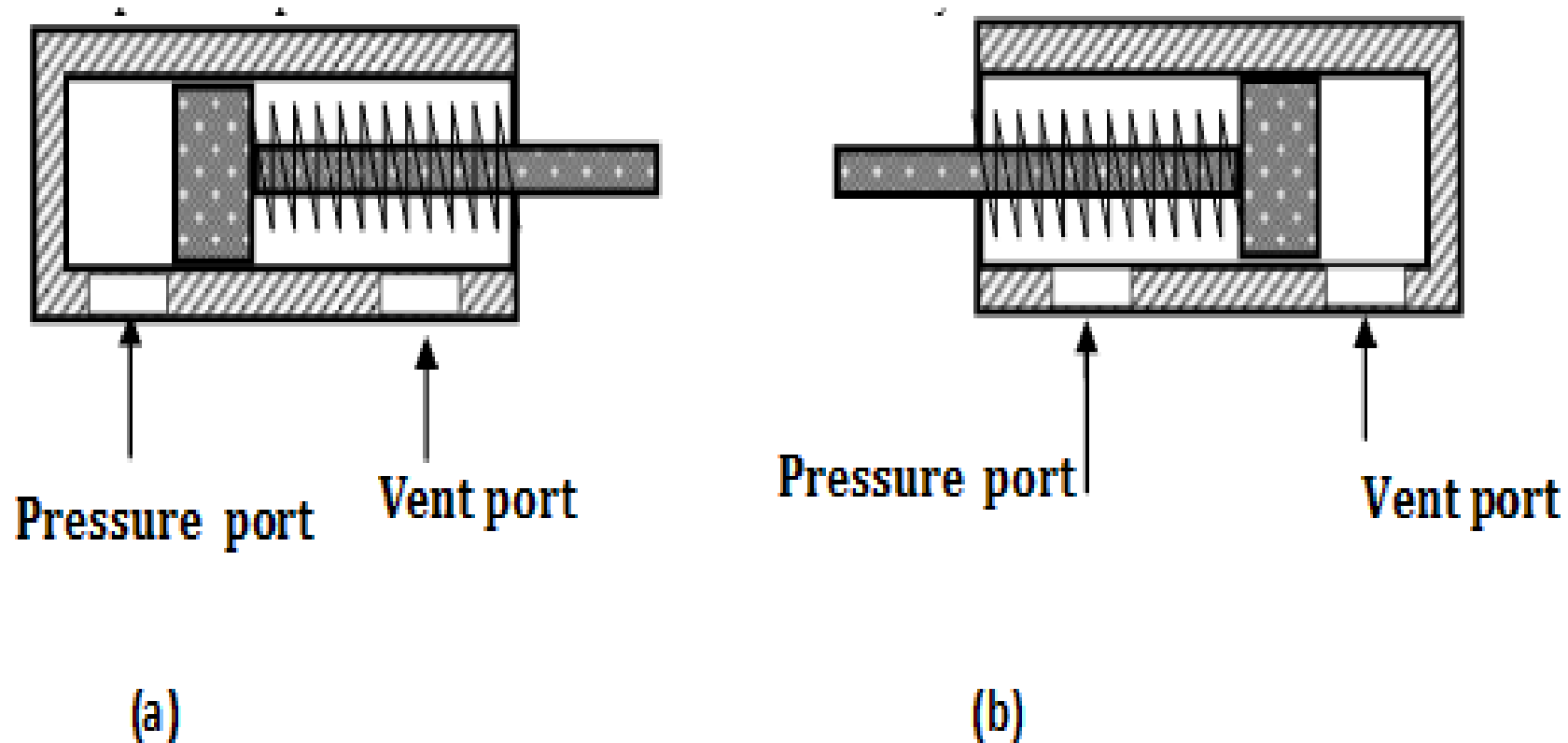


Figure 1.3 (a) Push- and (b) pull-type single-acting cylinders

# Double-Acting Cylinder:

There are two types of double-acting cylinders:

- **Double-acting cylinder with a piston rod on one side:**
- Double-acting cylinder with a piston rod on both sides.

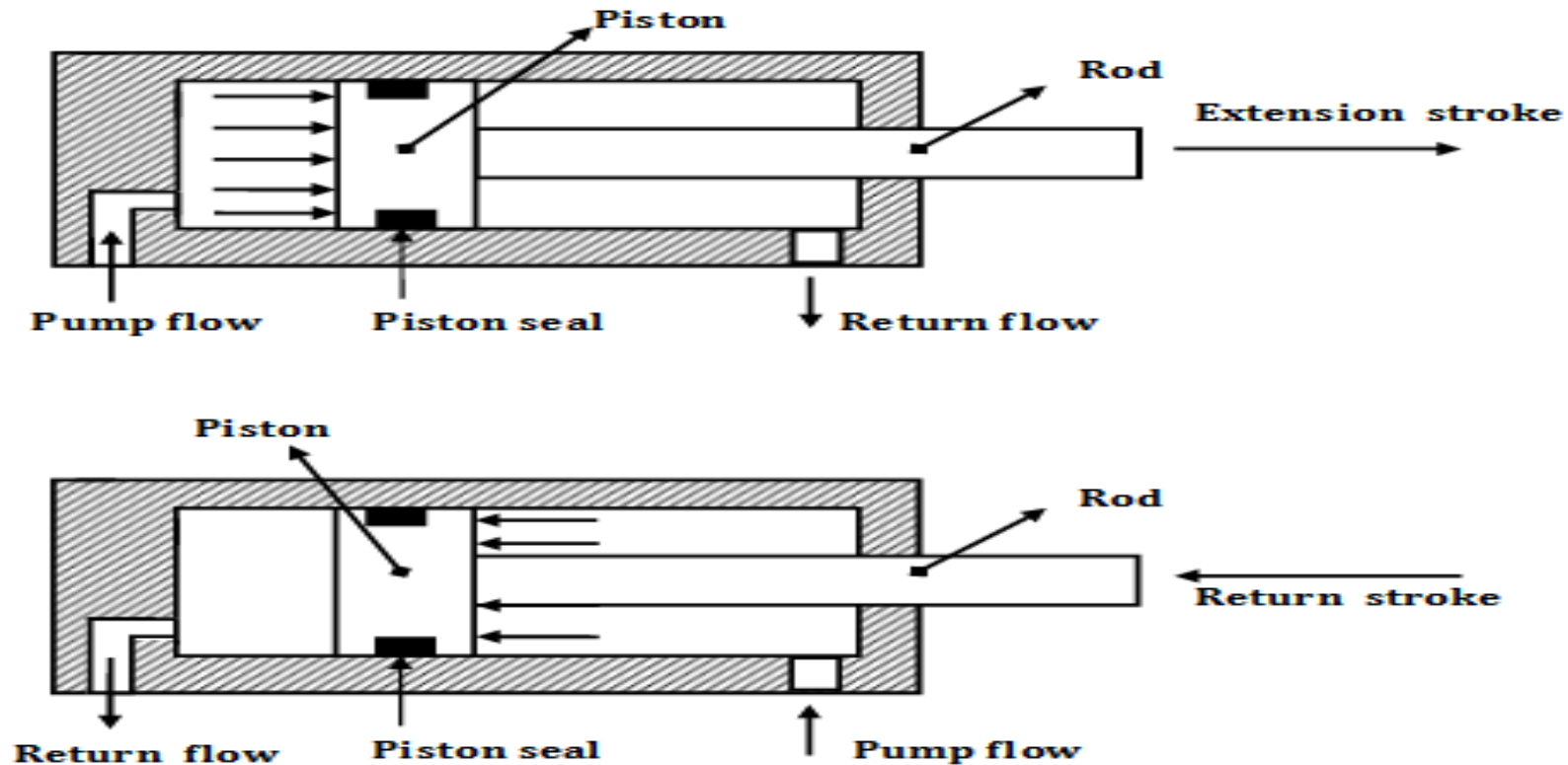


Figure 1.4 Double-acting cylinder with a piston rod on one side

## Double-acting cylinder with a piston rod on both sides:

- This cylinder can be used in an application *where work can be done by both ends of the cylinder*, thereby making the cylinder more productive.

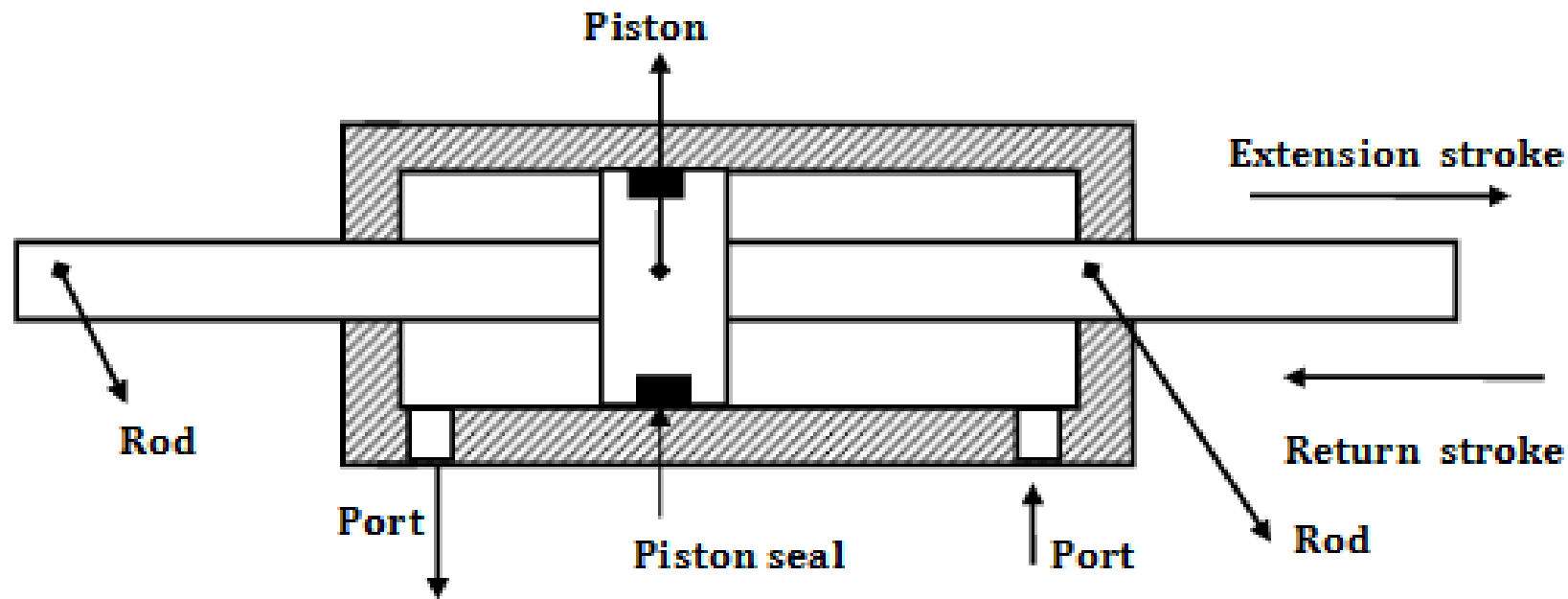


Figure 1.5 Double-acting cylinder with a piston rod on one side

# Telescopic Cylinder:

- A telescopic cylinder is used when a long stroke length and a short retracted length are required.
- *These cylinders have multiple stages that extend one out of the other, providing a longer stroke length than a single-stage cylinder.*
- The telescopic cylinder extends in stages, each stage consisting of a sleeve that fits inside the previous stage.
- They generally consist of a nest of tubes and operate on the displacement principle.
- The tubes are supported by bearing rings, the innermost (rear) set of which have grooves or channels to allow fluid flow.
- The front bearing assembly on each section includes seals and wiper rings. Stop rings limit the movement of each section, thus preventing separation.

# Telescopic Cylinder:

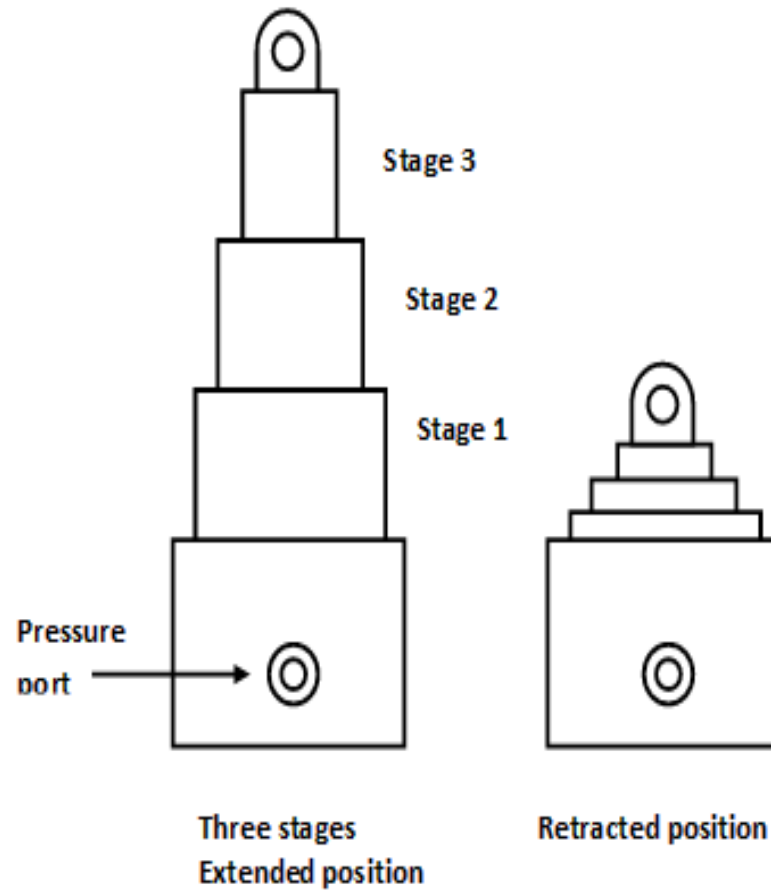
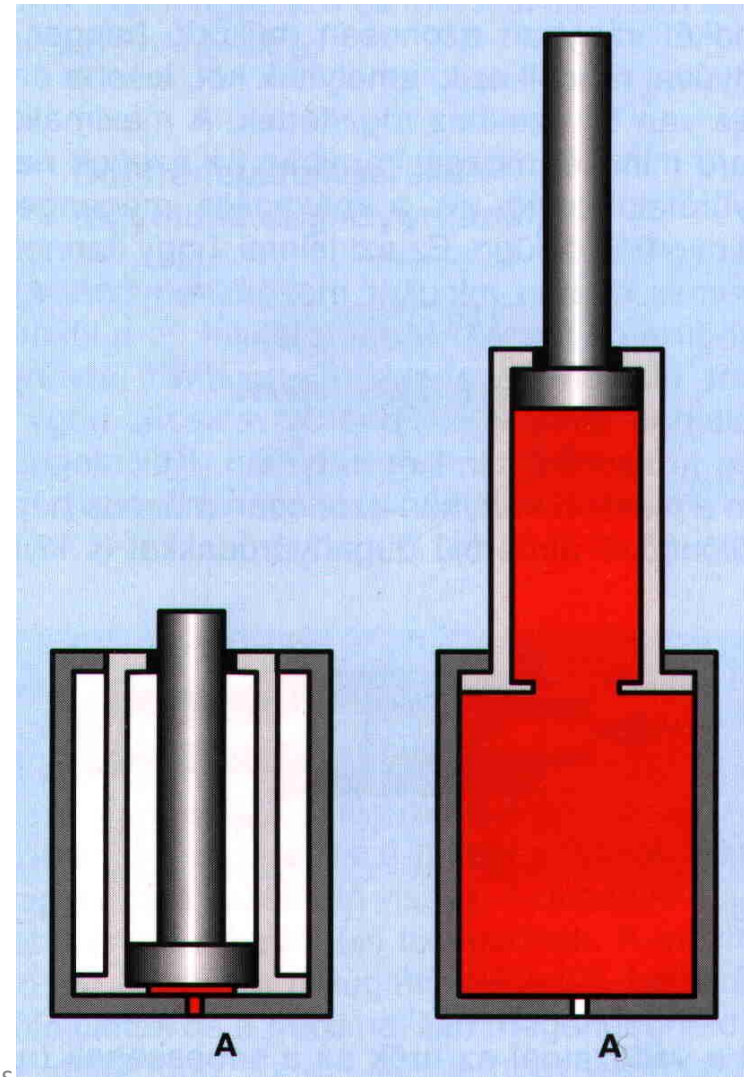


Figure 1.6 Telescopic cylinder



# Tandem Cylinder:

- A tandem cylinder is used in applications where a large amount of force is required from a small-diameter cylinder.
- *These cylinders consist of two or more single-acting cylinders plumbed together to provide increased force or stroke length.*
- They are a good option for applications where a single cylinder cannot provide the required force or stroke.
- Pressure is applied to both pistons, resulting in increased force because of the larger area.
- The drawback is that these cylinders must be longer than a standard cylinder to achieve an equal speed because flow must go to both pistons.

# Tandem Cylinder:

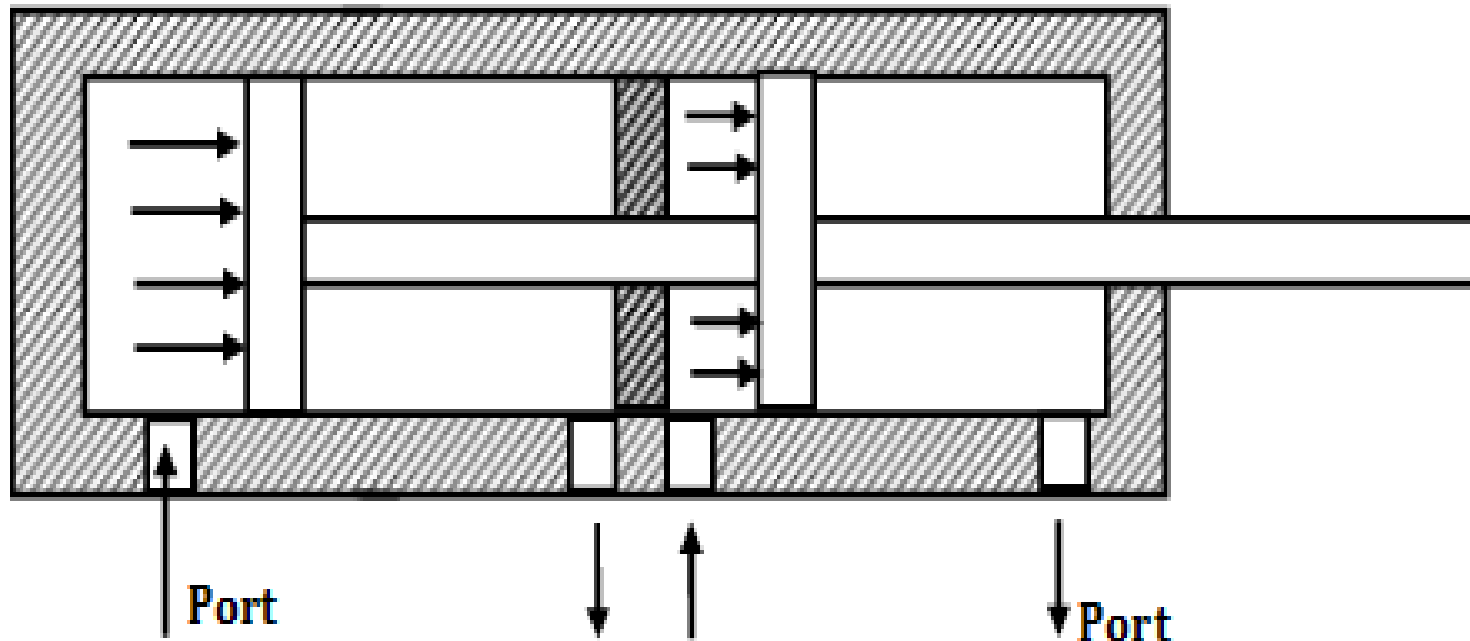


Figure 1.7 Tandem cylinder

Valves

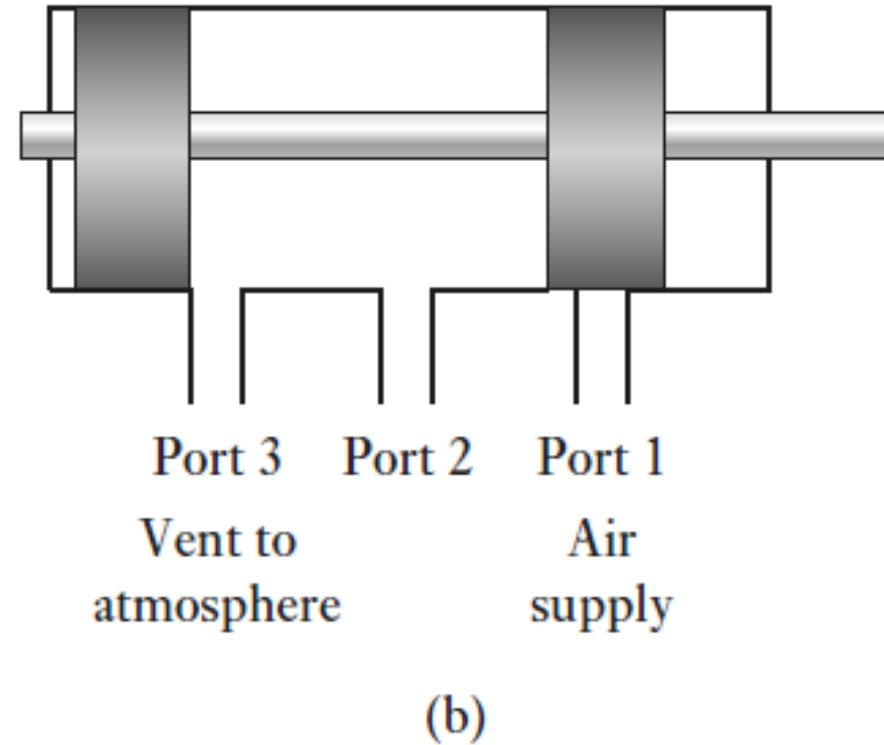
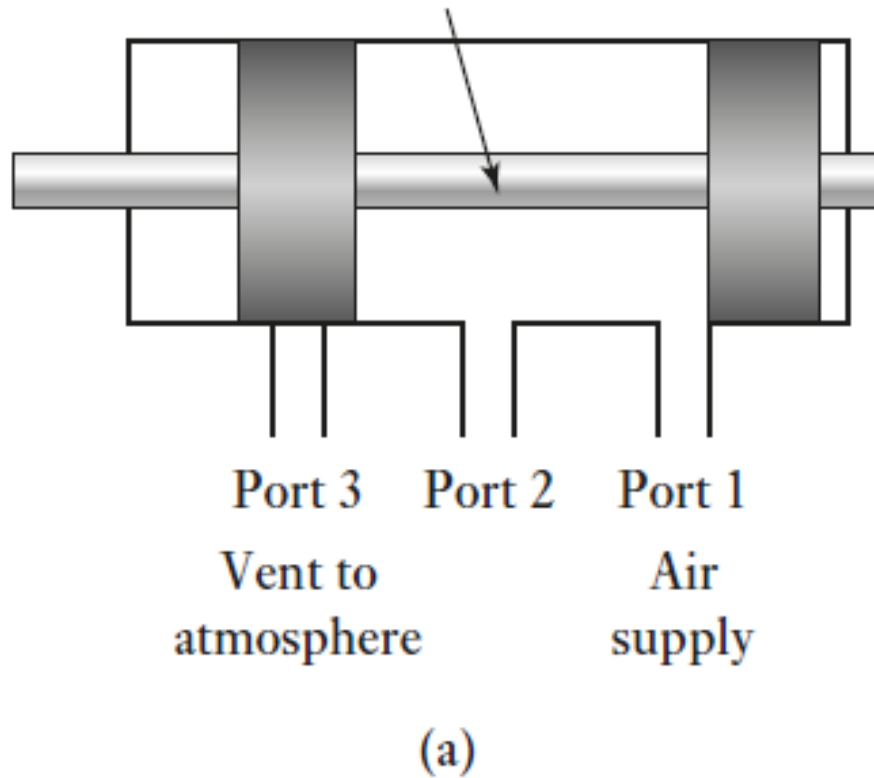
# Valves

- Valves are used with hydraulic and pneumatic systems to direct and regulate the fluid flow
- Two forms of valve
  - Finite position valves - to allow or block fluid flow
  - Infinite position valves - able to control flow anywhere between fully on and fully off

# Directional control valves

- Pneumatic and hydraulic systems use directional control valves to *direct the flow of fluid through a system*
- They are not intended to vary the rate of flow of fluid but are either completely open or completely closed
- They might be activated to switch the fluid flow direction by means of mechanical, electrical or fluid pressure signals

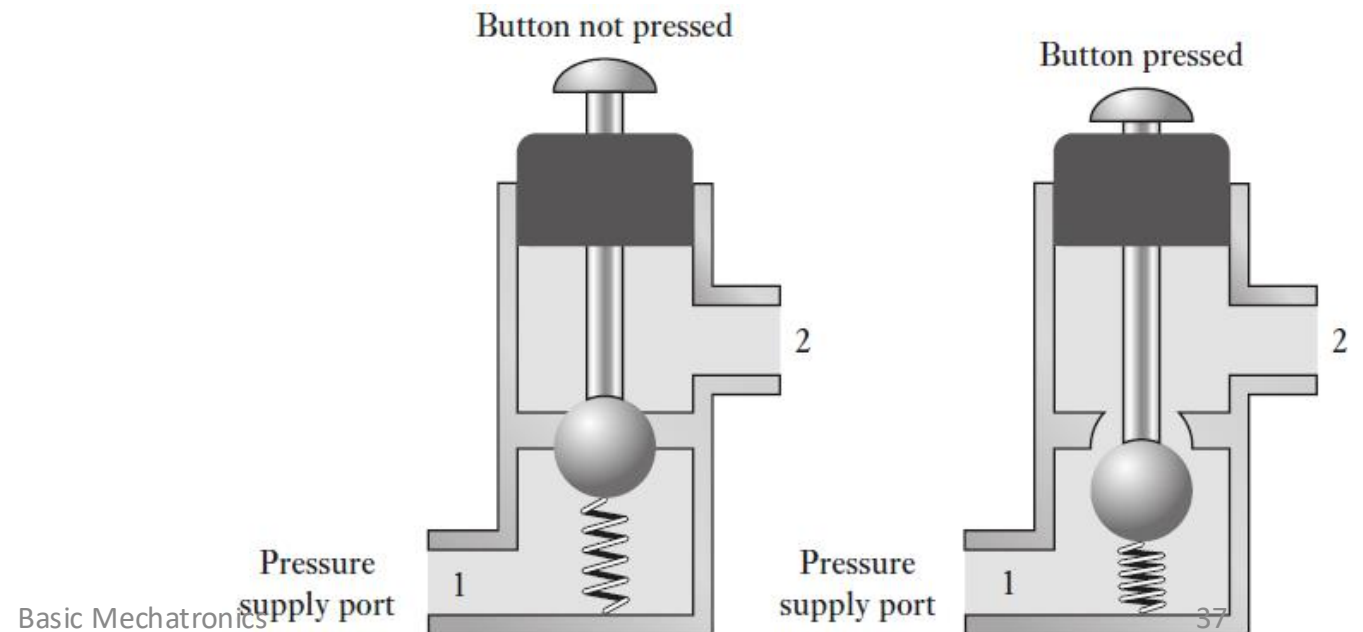
- Spool



- The air supply is connected to port 1 and port 3 is closed, the device connected to port 2 can be pressurised
- When the spool is moved to the left the air supply is cut off and port 2 is connected to port 3. Port 3 is a vent to the atmosphere and so the air pressure in the system attached to port 2 is vented

# Poppet valve - directional control valve

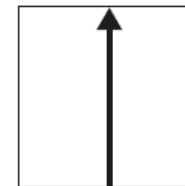
- Balls in conjunction with valve seats to control the flow.
- This valve is normally in the closed condition
- When the push-button is pressed, the ball is pushed out of its seat and flow occurs from port 1 to port 2.
- When the button is released, the spring forces the ball back up against its seat and so closes off the flow.



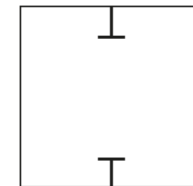
# Valve symbols

- a two-position valve will have two squares, a three-position valve three squares.
- Arrow-headed lines are used to indicate the directions of flow in each of the positions
- Blocked-off lines indicating closed flow lines
- Ports are labelled by a number or a letter according to their function.
  - 1 (or P) for pressure supply
  - 3 (or T) for hydraulic return port
  - 3 or 5 (or R or S) for pneumatic exhaust ports
  - 2 or 5 (or B or A) for output ports

**Figure 7.7** (a) Flow path, (b) flow shut-off, (c) initial connections.



(a)

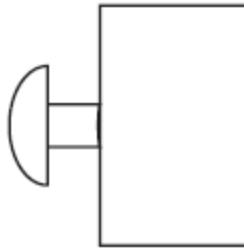


(b)

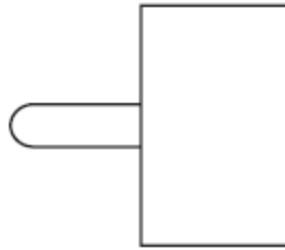


(c)

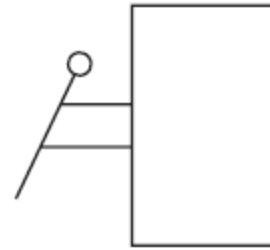
# Valve actuation symbols



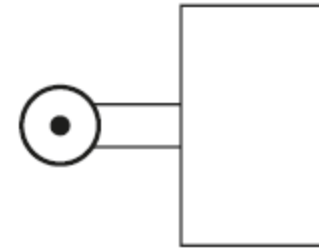
Push-button



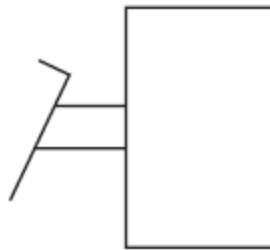
By plunger



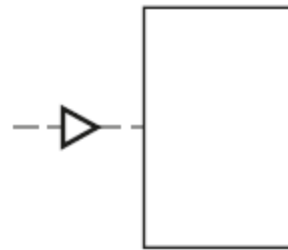
By lever



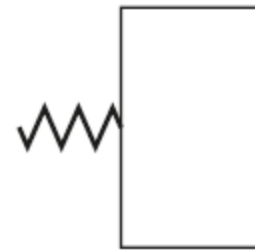
By roller



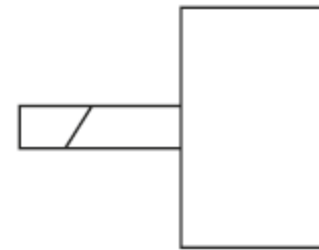
By pedal



By application of  
pneumatic pressure



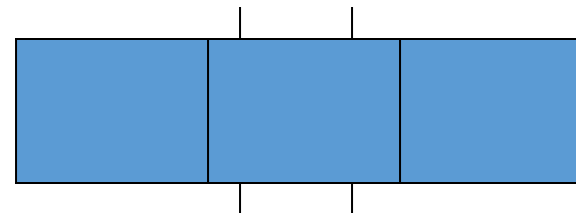
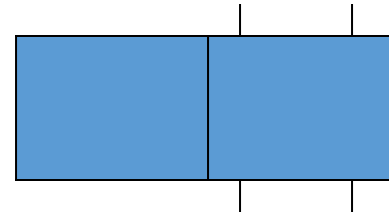
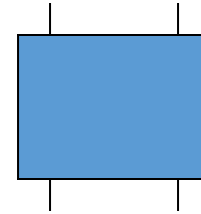
By spring



By solenoid

# Graphical representation

- PIPE CONNECTIONS
- Single position (initial)
- Two position valve
- Three position valve



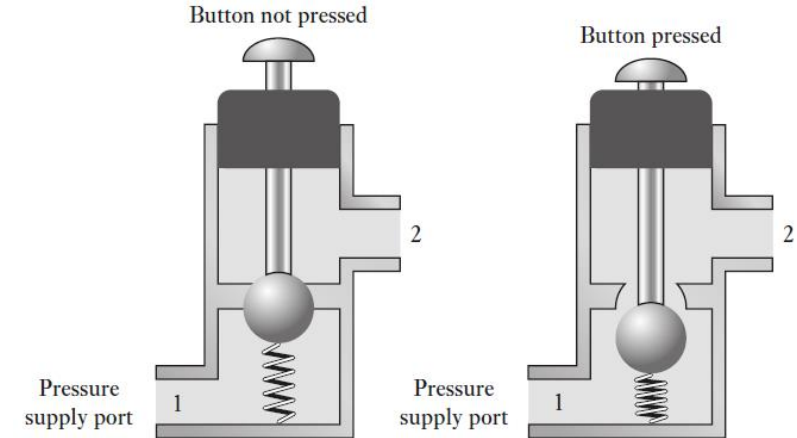
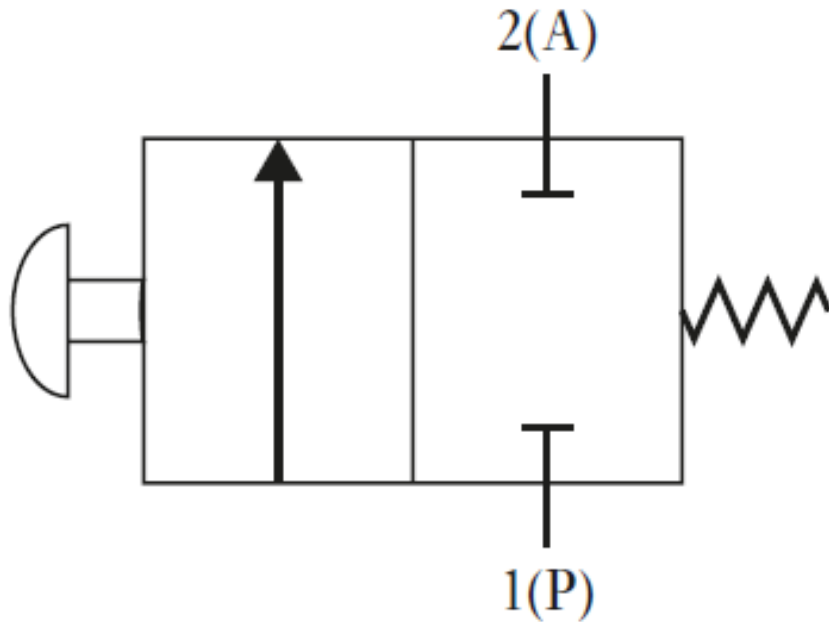
# Graphical representation

- PORT & POSITIONS TABLE:

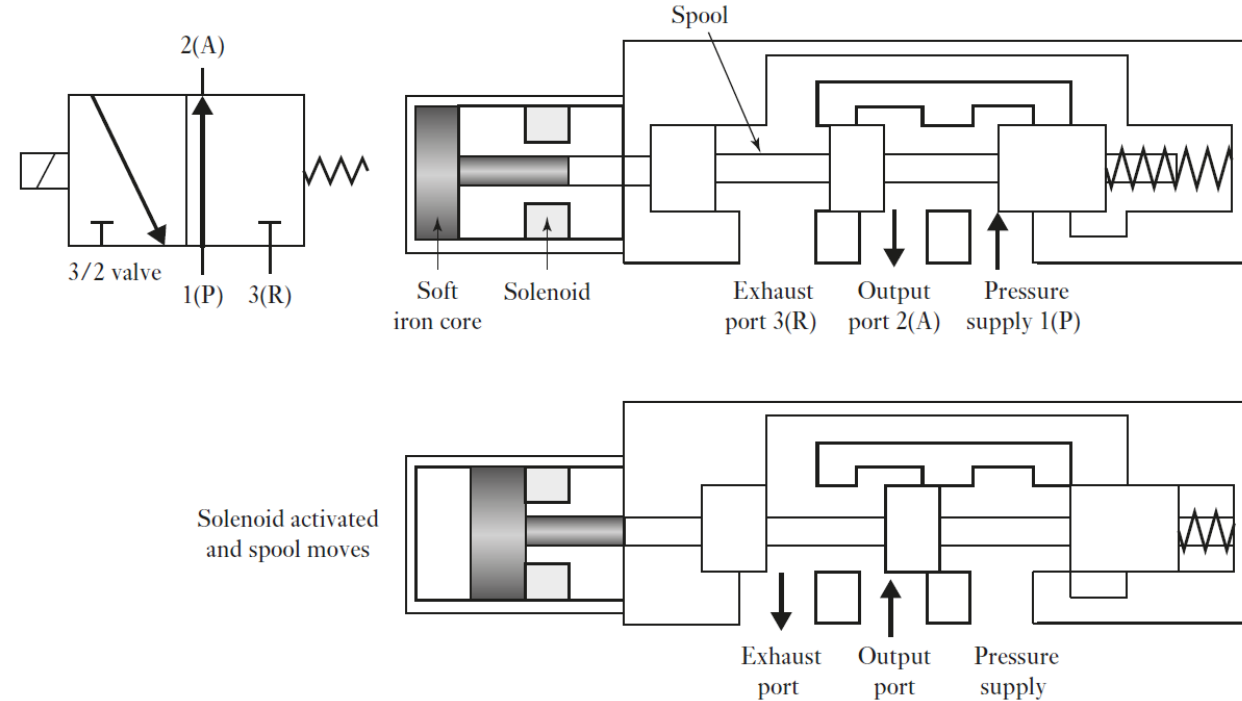
| Port                | Alphabetic system | No. sys. | Comment                            |
|---------------------|-------------------|----------|------------------------------------|
| Pressure port       | P                 | 1        | Supply port                        |
| Working port        | A                 | 2        | 3/2 DC valve                       |
| Working ports       | A, B              | 4, 2     | 4/2 or 5/2 DC valve                |
| Exhaust port        | R                 | 3        | 3/2 DC valve                       |
| Exhaust ports       | R, E,S            | 3, 5     | 3/2 DC valve                       |
| Pilot port          | Z or Y            | 12       | Pilot line (1 $\longrightarrow$ 2) |
| Pilot port          | Z                 | 14       | Pilot line (1 $\longrightarrow$ 4) |
| Pilot port          | Z or Y            | 10       | Pilot line (flow closed)           |
| Internal pilot port | Pz, Py            | 81, 91   | Auxiliary pilot air                |

# 2/2 Valve

- The two-port, two-position poppet valve described as 2/2 valve, the first number indicating the number of ports and the second number the number of positions.
- The valve actuation is by a push-button and a spring



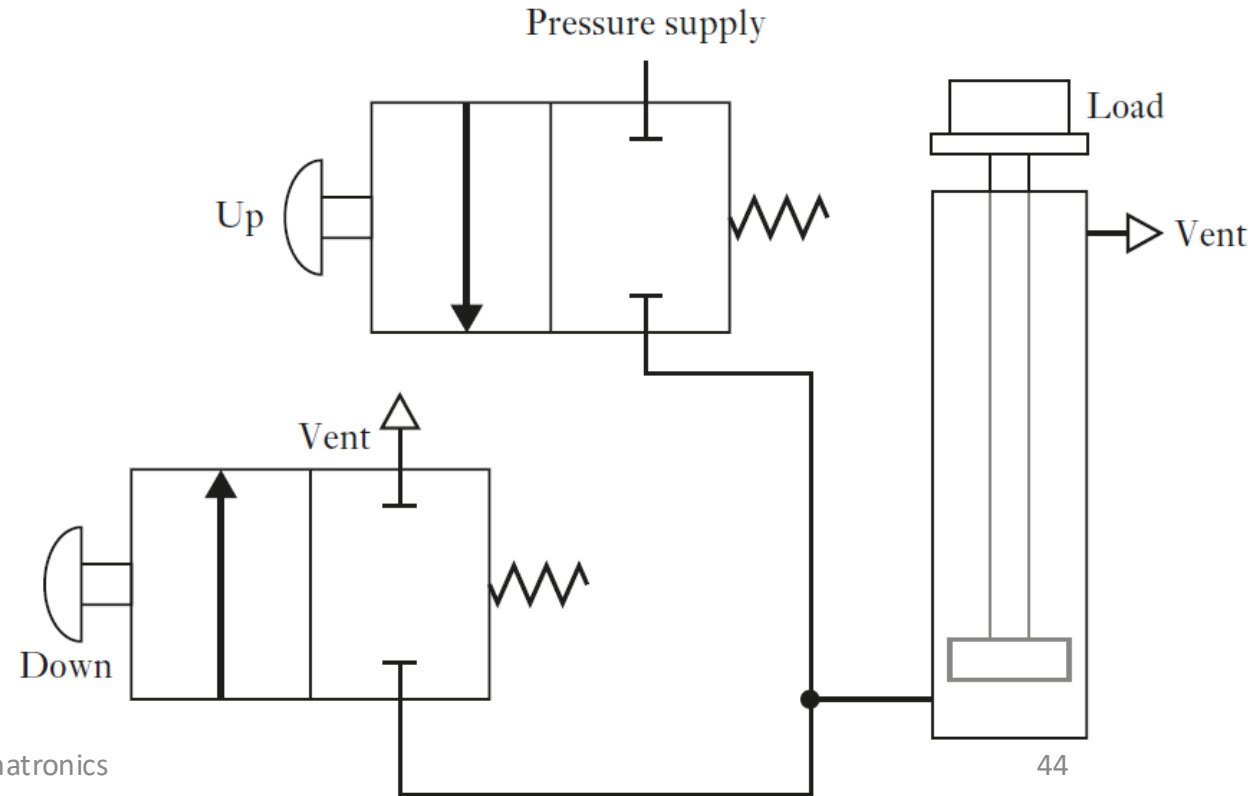
# 3/2 Valve (solenoid-operated spool valve)



- The valve is actuated by a current passing through a solenoid and returned to its original position by a spring
- 1(P) is connected to 2(A) and 3(R) closed.
- When the solenoid is activated 1(P) closed and 2(A) connected to 3(R)
- When the current through the solenoid ceases, the spring pushes the valve back to its initial position.

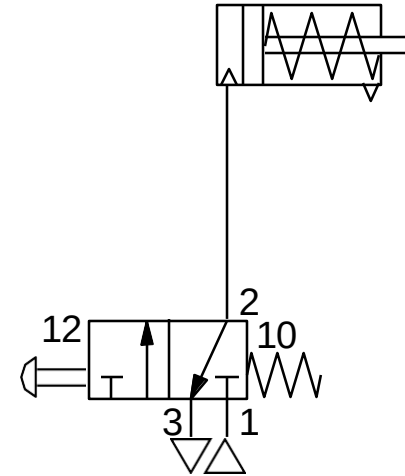
# Lift system

- Two push-button 2/2 valves are used.
- When the button on the up valve is pressed, the load is lifted.
- When the button on the down valve is pressed, the load is lowered.
- open arrow is used to indicate a vent to the atmosphere



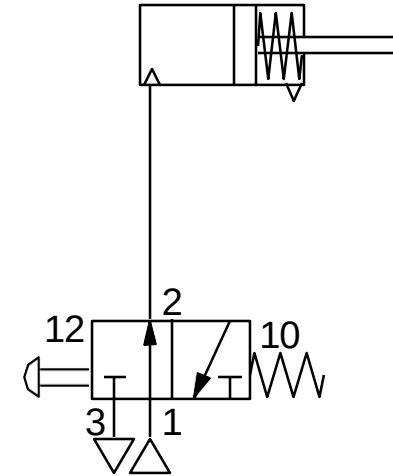
# 3/2 valve actuator control

- A 3 port valve provides the inlet and exhaust path and is the normal choice for the control of a single acting cylinder
- In the normal position produced by the spring, the valve is closed
- In the operated position produced by the push button the valve is open
- The push button must be held down for as long as the cylinder is outstroked



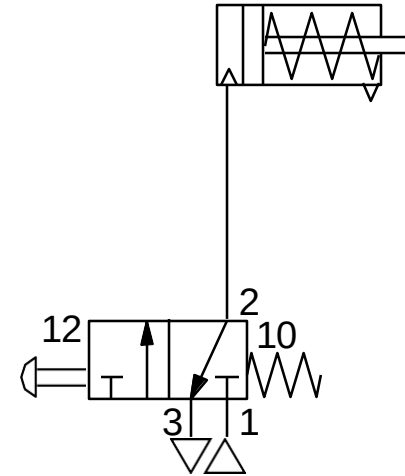
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- In the normal position produced by the spring, the valve is closed
- In the operated position produced by the push button the valve is open
- The push button must be held down for as long as the cylinder is outstroked

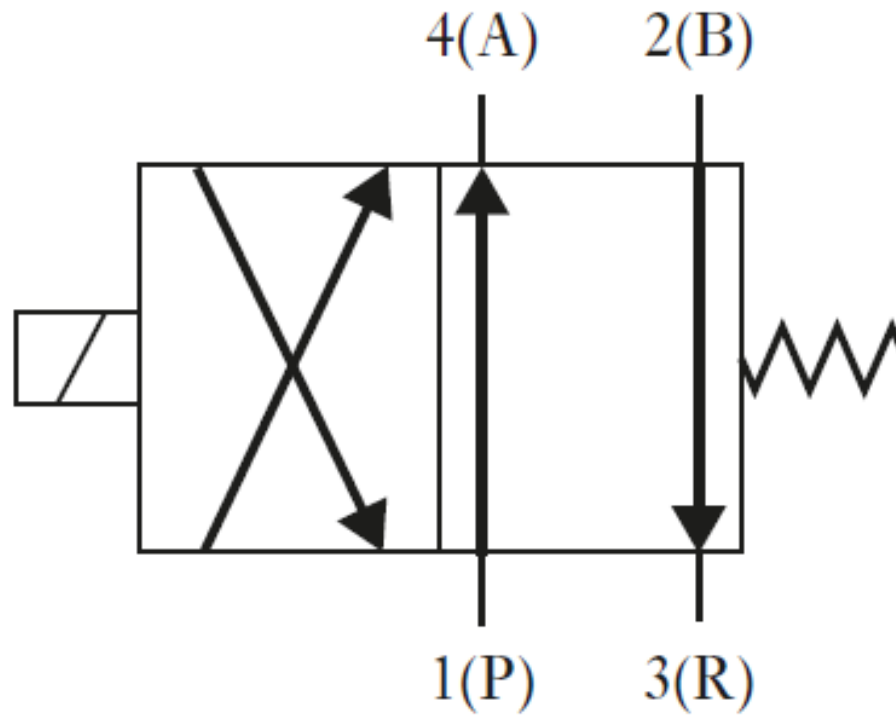


# 3/2 valve actuator control

- A 3 port valve provides the inlet and exhaust path and is the normal choice for the control of a single acting cylinder
- In the normal position produced by the spring, the valve is closed
- In the operated position produced by the push button the valve is open
- The push button must be held down for as long as the cylinder is outstroked

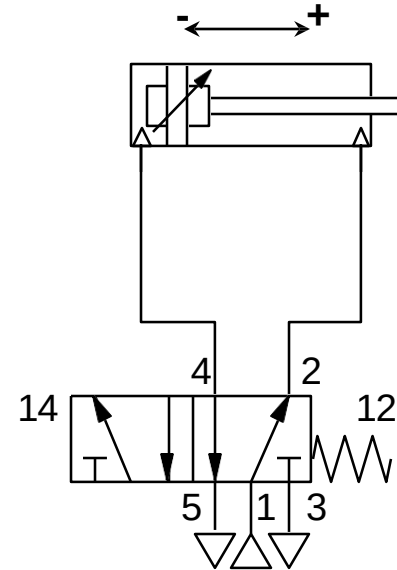


# 4/2 Valve

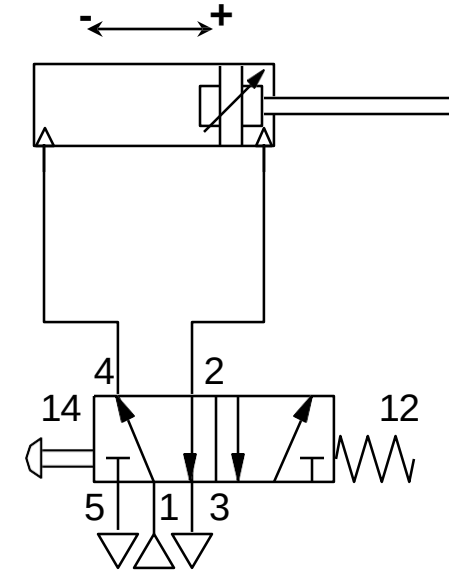


# 5/2 Valve actuator control

- For a double acting cylinder the power and exhaust paths are switched simultaneously
- When the button is pushed the supply at port 1 is connected to port 4 and the outlet port 2 connected to exhaust port 3. The cylinder moves plus
- When the button is released port 1 is connected to port 2 and port 4 connected to port 5. Cylinder minus



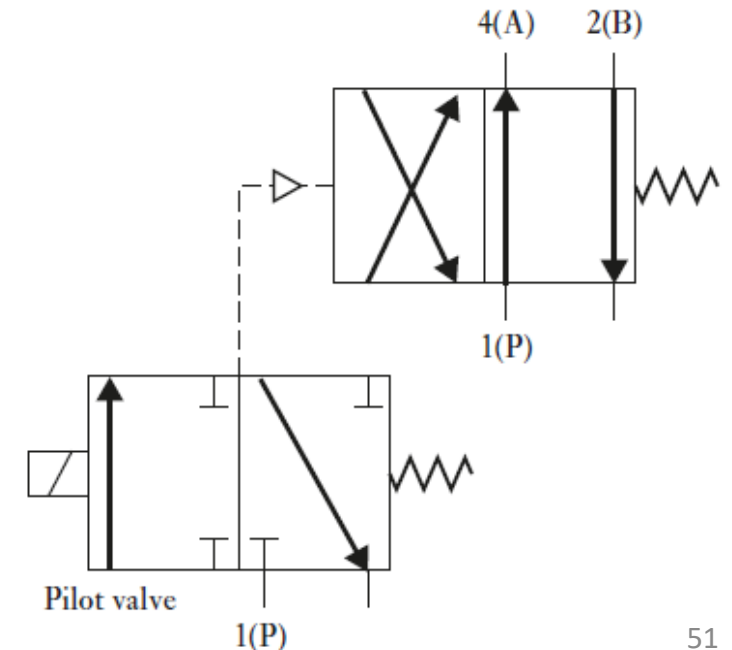
# 5/2 Valve actuator control



- For a double acting cylinder the power and exhaust paths are switched simultaneously
- When the button is pushed the supply at port 1 is connected to port 4 and the outlet port 2 connected to exhaust port 3. The cylinder moves plus
- When the button is released port 1 is connected to port 2 and port 4 connected to port 5. Cylinder minus

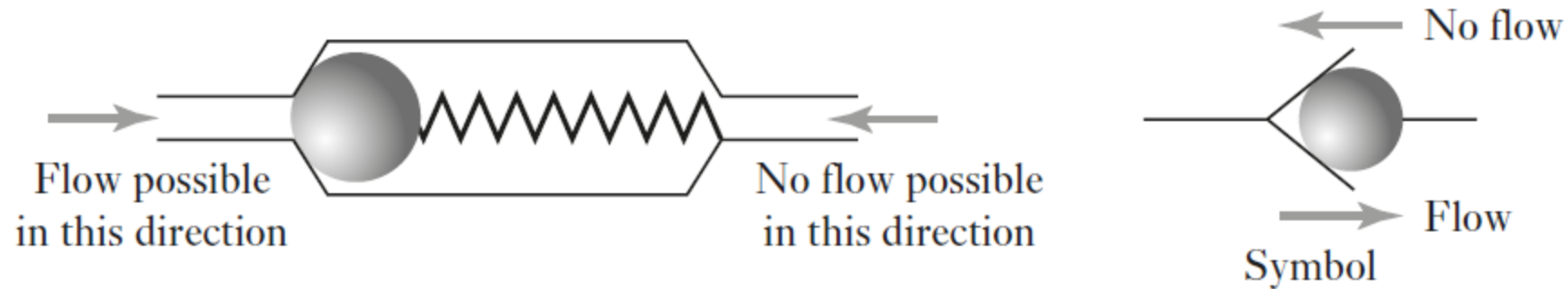
# Pilot-operated valves

- The force required to move the ball or shuttle in a valve can often be too large for manual or solenoid operation.
- To overcome this problem a pilot-operated system is used where one valve is used to control a second valve
- The pilot pressure line is indicated by dashes.
- The pilot and main valves can be operated by two separate valves but they are often combined in a single housing.



# Directional valves

- Free flow can only occur in one direction through the valve: that which results in the ball being pressed against the spring.
- Flow in the other direction is blocked by the spring forcing the ball against its seat



# Methods of actuation

## Manual

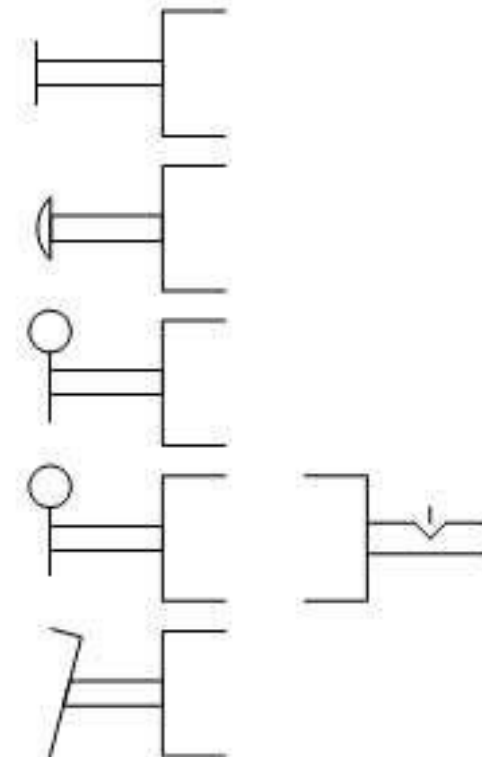
General

Pushbutton

Lever Operated

Detend lever operated

Foot pedal



# Methods of actuation

## Mechanical

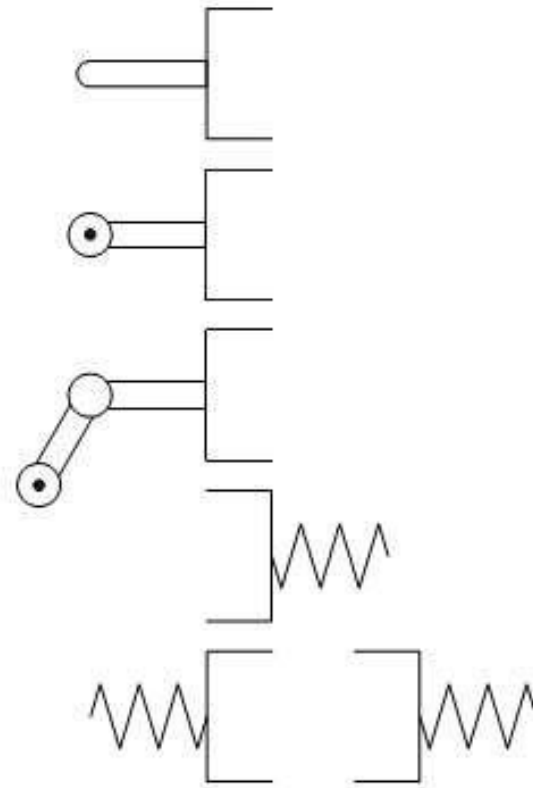
Plunger

Roller operated

Idle return, roller

Spring return

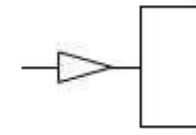
Spring centred



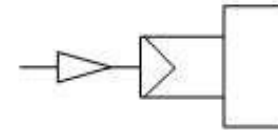
# Methods of actuation

## Pneumatic

Direct pneumatic actuation



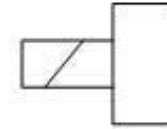
Indirect pneumatic actuation (piloted)



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## Electrical

Single solenoid operation



Double solenoid operation



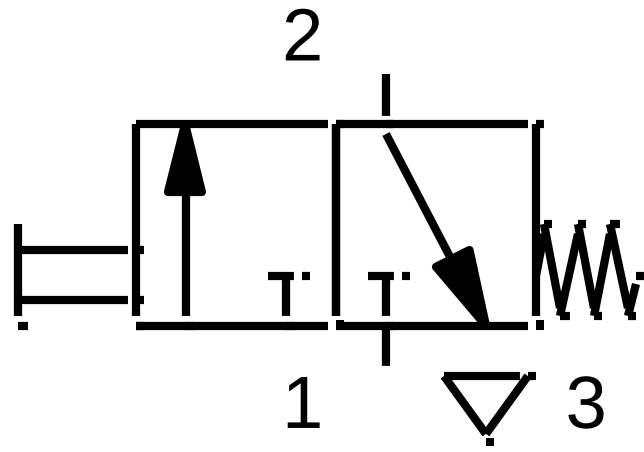
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## Combined

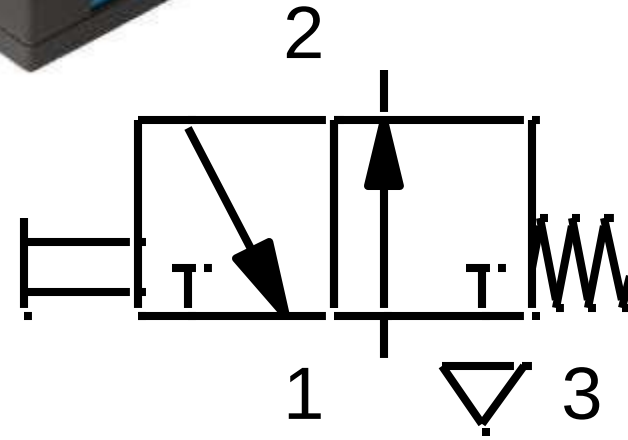
Double solenoid and pilot operation with manual override



# 3/2 way valve

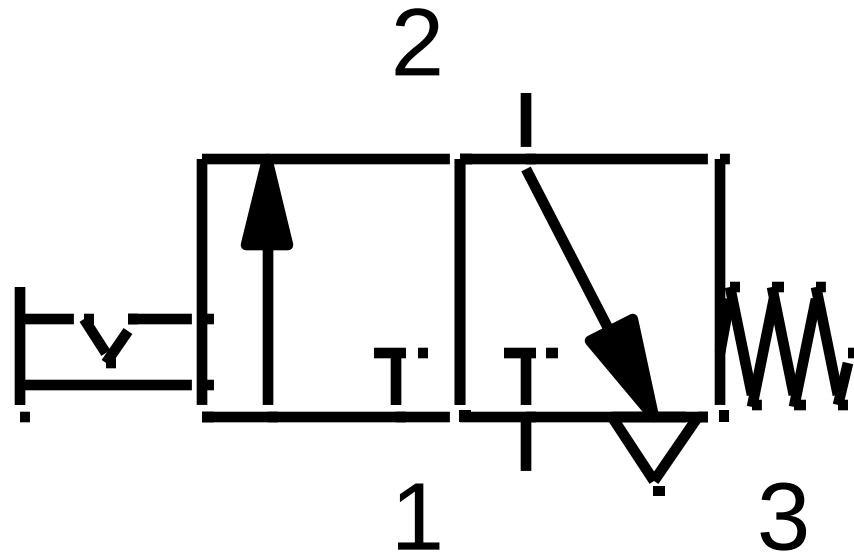


Push button operated spring returned  
3/2 way normally closed  
valve



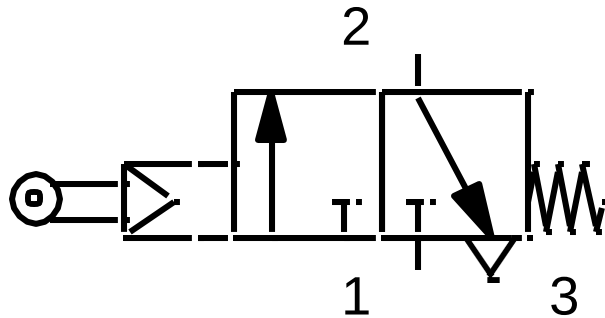
Push button operated spring returned  
3/2 way normally open  
valve

# Striking switch

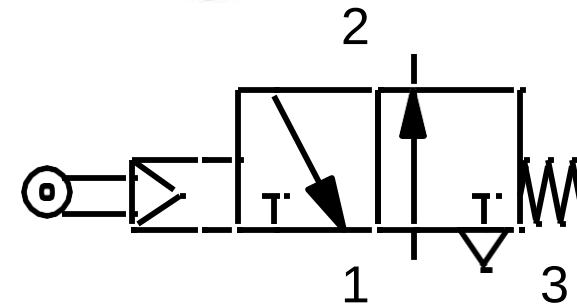


striking switch  
3/2 way normally closed valve

# 3/2 way valve

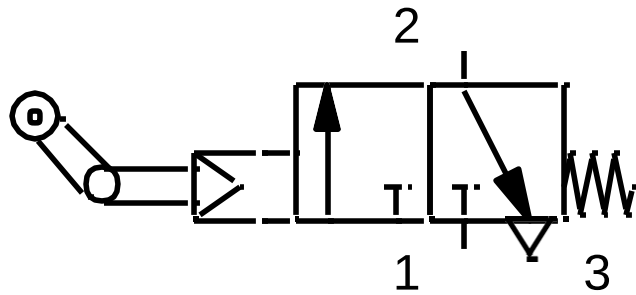


Roller operated spring returned  
3/2 way normally closed valve



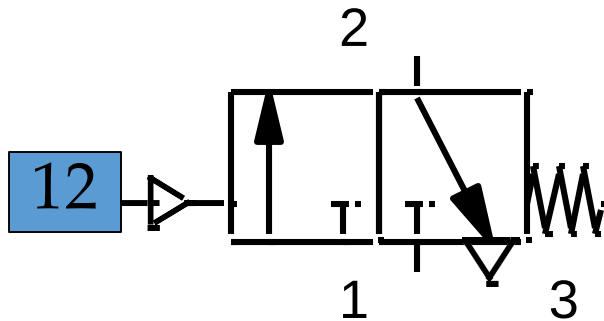
Roller operated spring returned  
3/2 way normally open valve

# 3/2 way Idle return roller operated spring returned

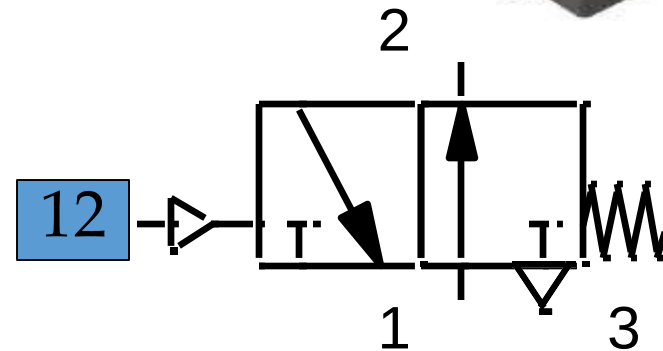


Idle return roller operated  
spring returned  
3/2 way normally Closed valve

# Pneumatically operated spring returned



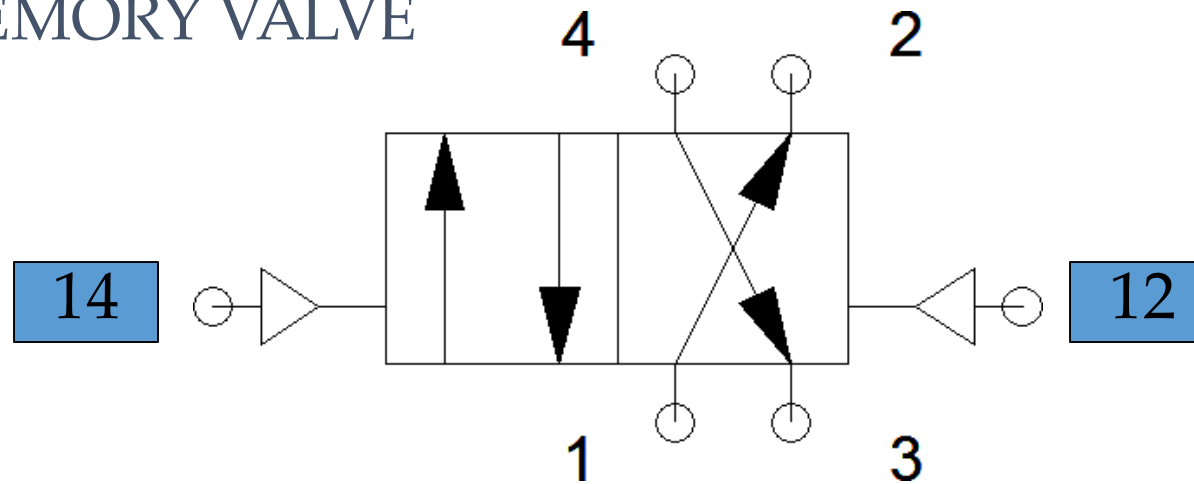
Pneumatically operated  
spring returned  
3/2 way normally closed valve  
(3/2 way single pilot)



Pneumatically operated  
spring returned  
3/2 way normally open valve  
(3/2 way single pilot)

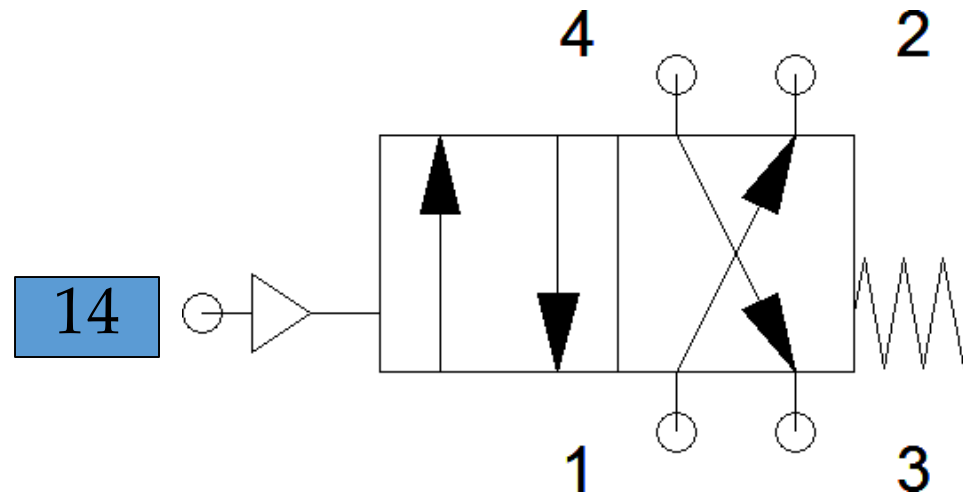
# 4/2 BOTH SIDE PILOT OPERATED VALVE

- MEMORY VALVE



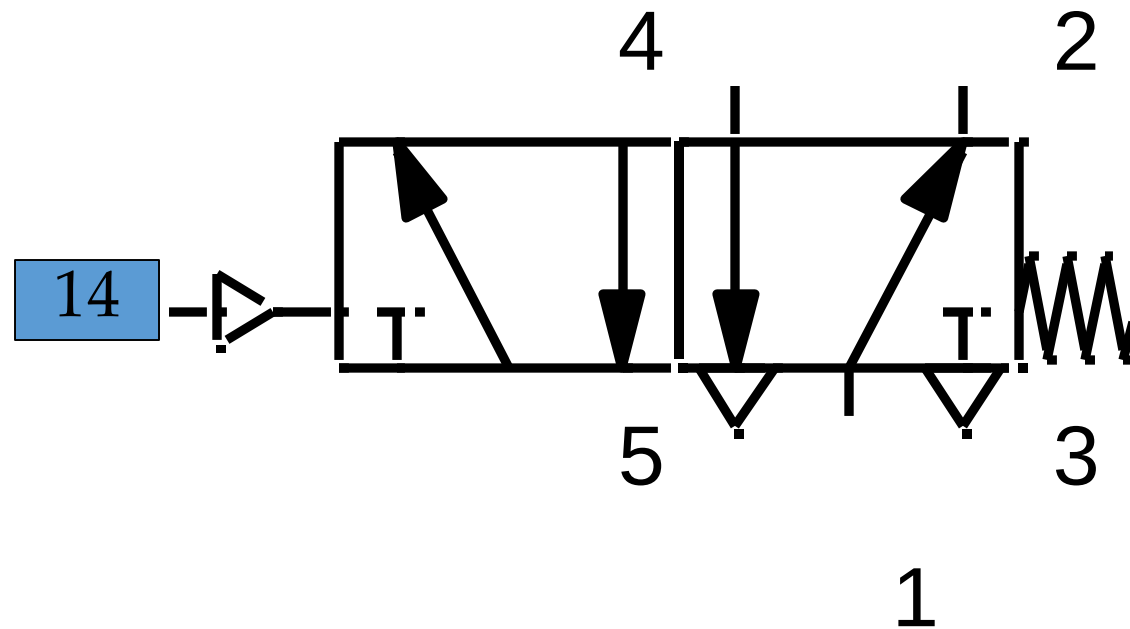
Pneumatically operated Spring returned 5/2 way valve (5/2 way single pilot)

# 4/2 SINGLE SIDE PILOT OPERATED SPRING TO RETURN



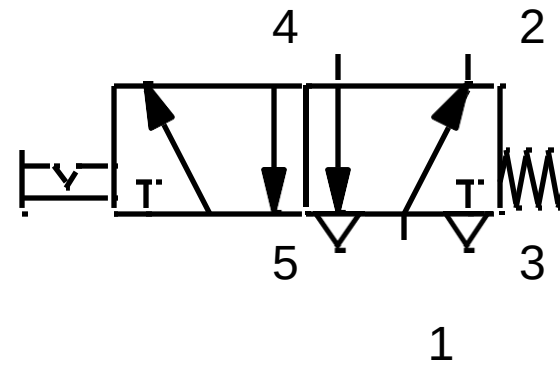
Pneumatically operated Spring returned 4/2 way valve (4/2 way single pilot)

# Pneumatically operated Spring returned



Pneumatically operated Spring returned 5/2 way valve  
(5/2 way single pilot)

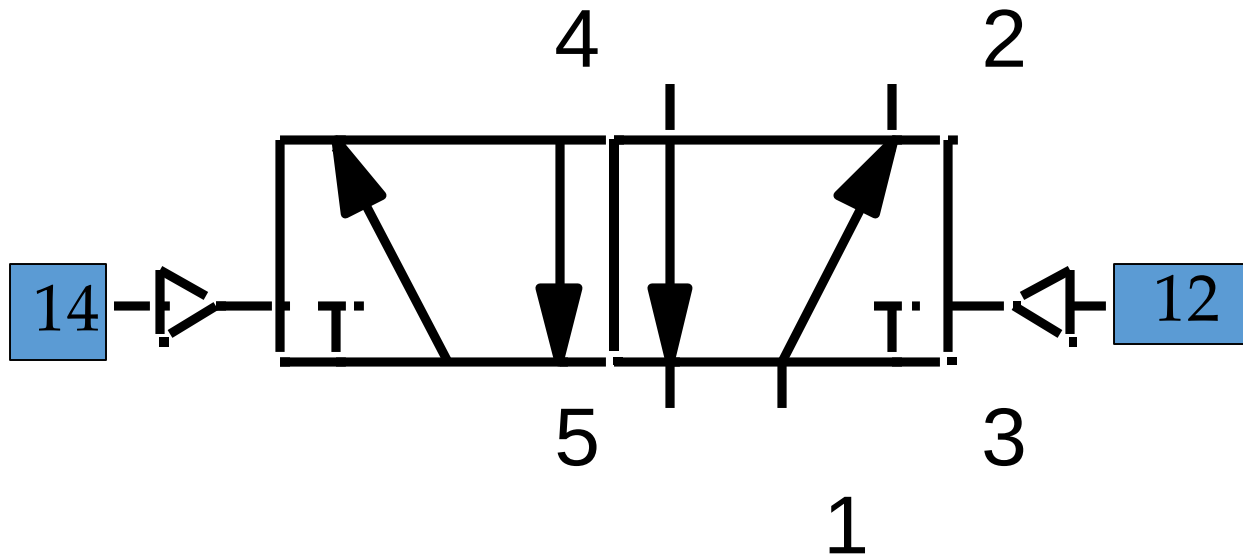
# Push button operated Spring returned



striking switch 5/2 way valve

# Both pneumatically operated valve

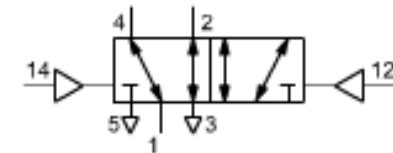
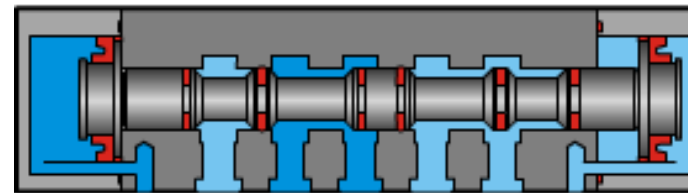
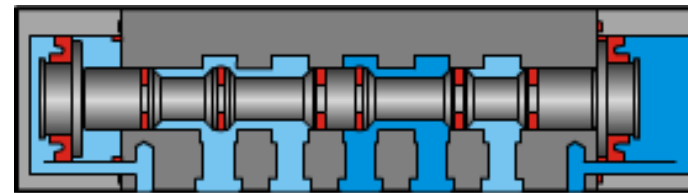
- 5/2 way valve



Pneumatically operated pneumatically  
return 5/2 way valve  
(5/2 way double pilot valve / memory valve)

# 5/2 way double pilot valve

5/2 way  
double pilot  
valve with  
longitudinal  
slide



# Pressure control valves

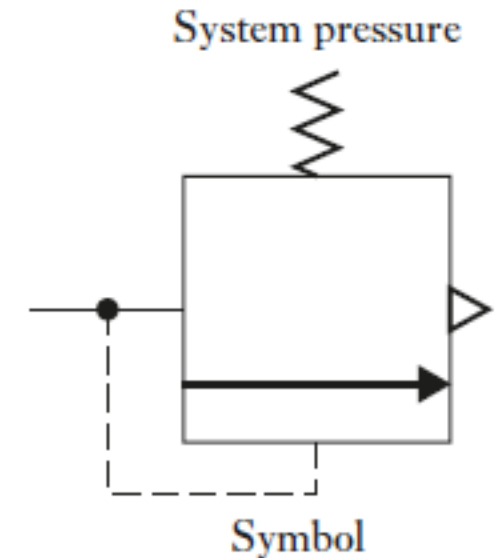
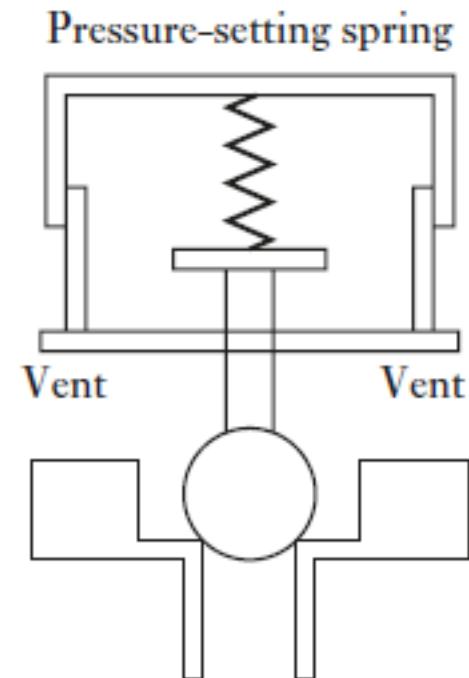
- adjust or regulate the pressure for system component or end users.

It can performs

- Limit system pressure
- Reduce pressures
- Set pressure at which oil enters a circuit
- Unload a pump

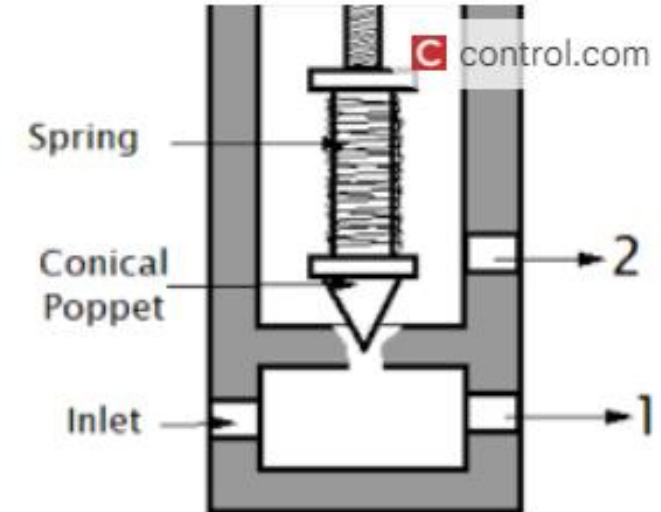
1. Pressure regulating valves - These are used to control the operating pressure in a circuit and maintain at a constant value.

2. Pressure limiting valves - These are used as safety devices to limit the pressure in a circuit to below some safe value.
- The valve opens and vents to the atmosphere, or back to the sump, if the pressure rises above the set safe value.
  - When the **inlet pressure overcomes the force exerted by the spring, the valve opens and vents** to the atmosphere, or back to the sump.



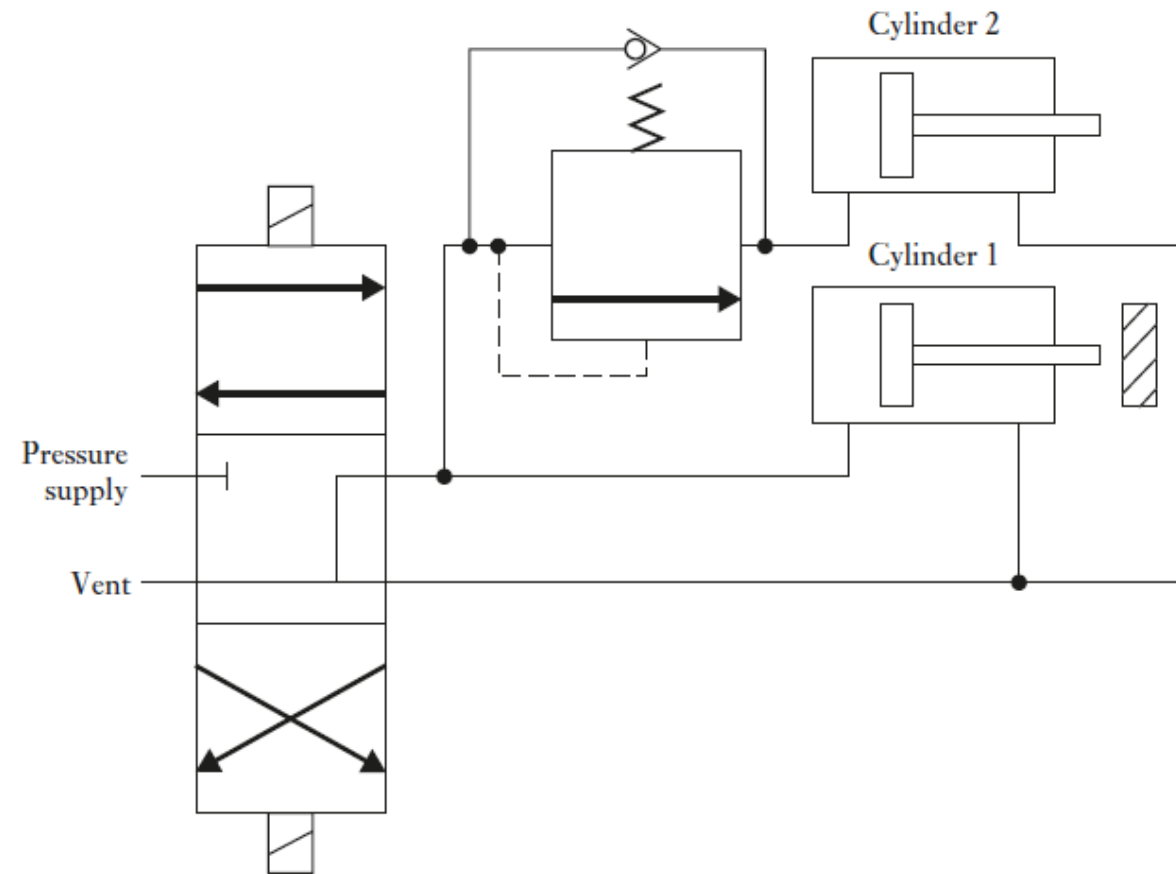
### 3. Pressure sequence valves / sequencing valve

- Its *primary function is to control the order of operations* in the system based on pressure
  - These valves are used to *sense the pressure of an external line and give a signal when it reaches some preset value.*
- The valve remains closed until the pressure at the inlet port reaches a preset level.
  - Once the pressure reaches the setpoint, the valve opens, allowing fluid to flow to a secondary circuit.
  - This sequential activation ensures that specific actions occur in a predetermined order.



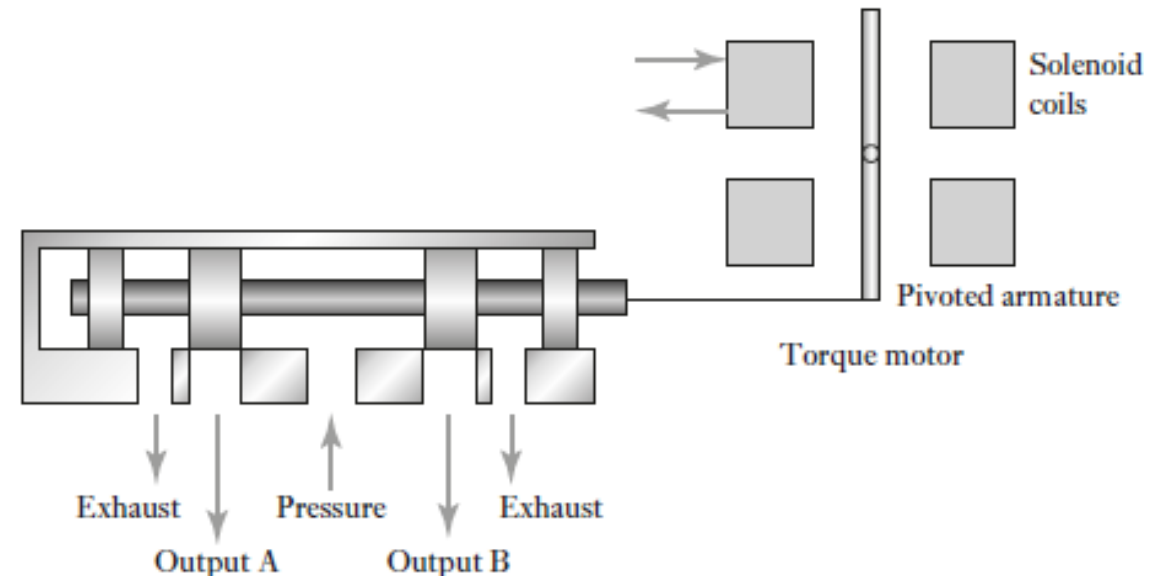
## Pressure sequence valves - Example

- When the 4/3 valve first operates, the pressure is applied to cylinder 1 and its ram moves to the right - While this is happening the pressure is too low to operate the sequence valve and so no pressure is applied to cylinder 2
- When the ram of cylinder 1 reaches the end stop, then the pressure in the system rises and, at an appropriate level, triggers the sequence valve to open and so apply pressure to cylinder 2 to start its ram in motion.



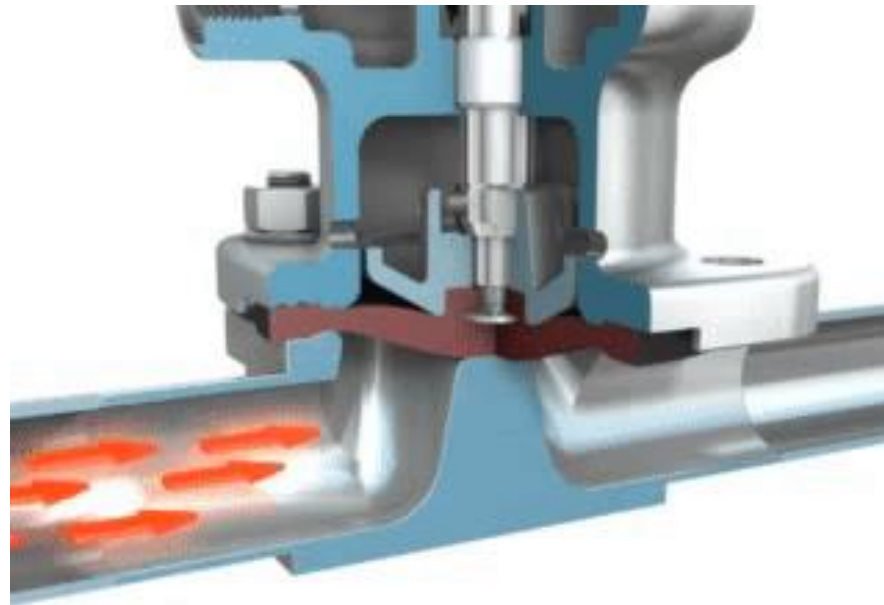
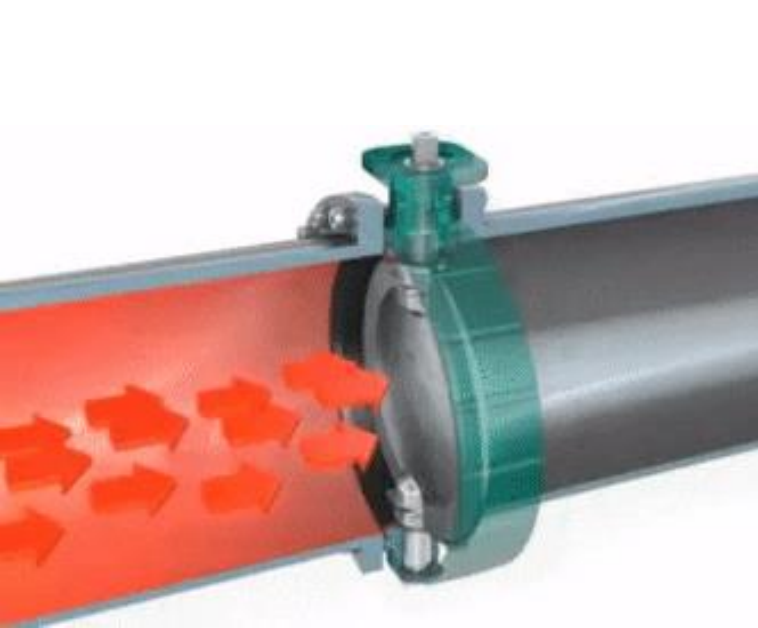
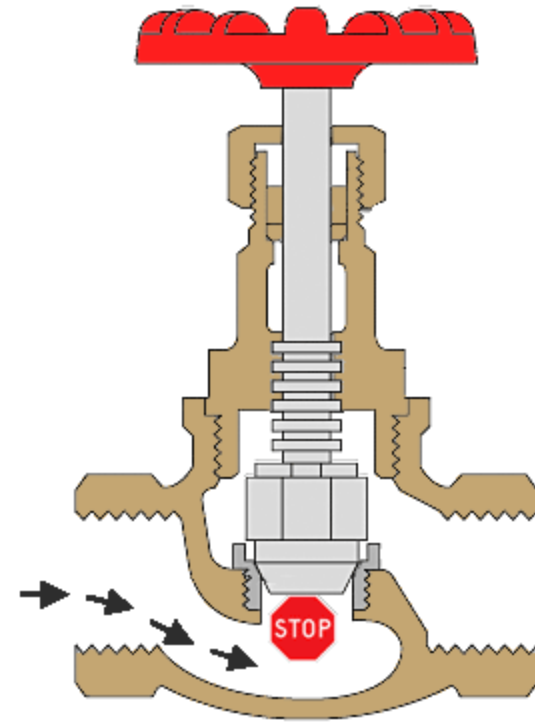
# Servo and Proportional control valves

- Infinite position valves
- *Spool displacement proportional to the current supplied to a solenoid*
- Servo valves have a torque motor to move the spool within a valve
  - By varying the current supplied to the torque motor, moves the spool in the valve and hence gives a flow related to the current.
  - Servo valves are high precision and costly and generally used in a closed-loop control system.
- Proportional control valve - position of the spool is directly controlled by the size of the current to the valve solenoid.
  - Less expensive
  - often used in open-loop control systems



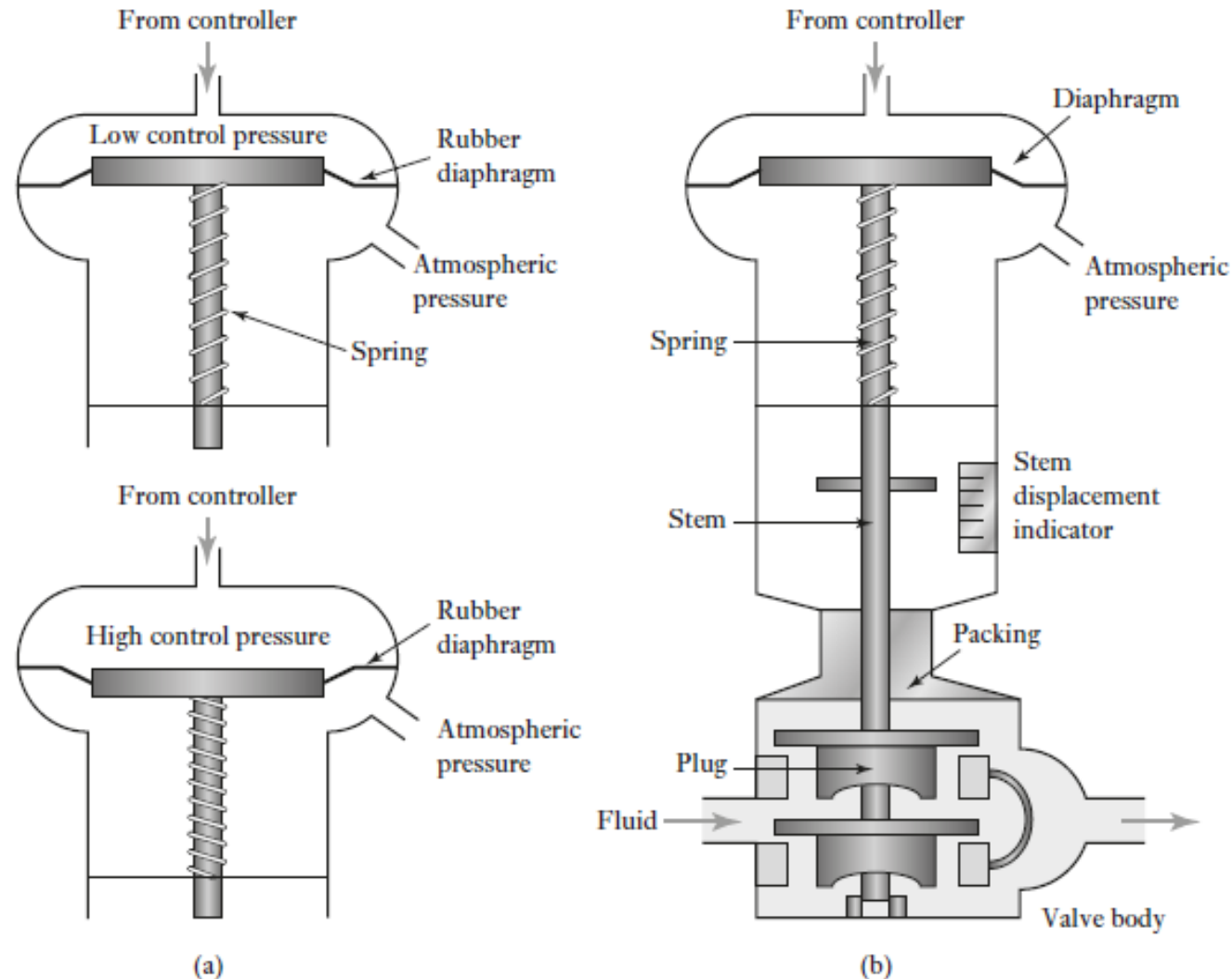
# Process control valves

- Control the rate of fluid flow
  - Butterfly valves - These valves use a disc that rotates to control flow.
  - Globe valves: These valves use a screw or plug that moves up and down to control flow
  - Ball valves: These valves use a ball with a hole in it to control flow.
  - Diaphragm valves: These valves use a flexible diaphragm to control flow.



- *The Process control valves is working based on an actuator being used to move a plug into the flow pipe and so alter the cross-section of the pipe through which the fluid can flow.*

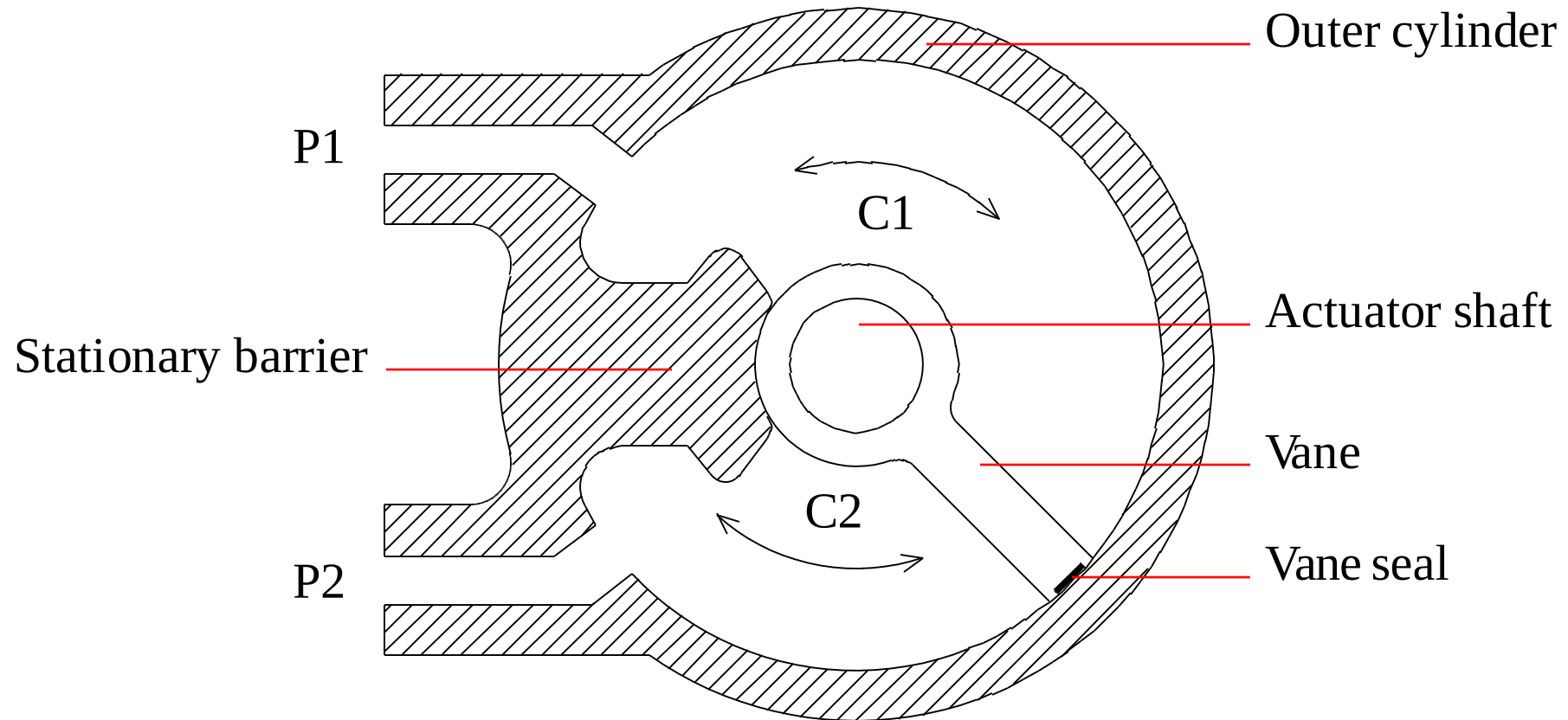
**Figure 7.23** (a) Pneumatic diaphragm actuator, (b) control valve.



# Rotary Actuators

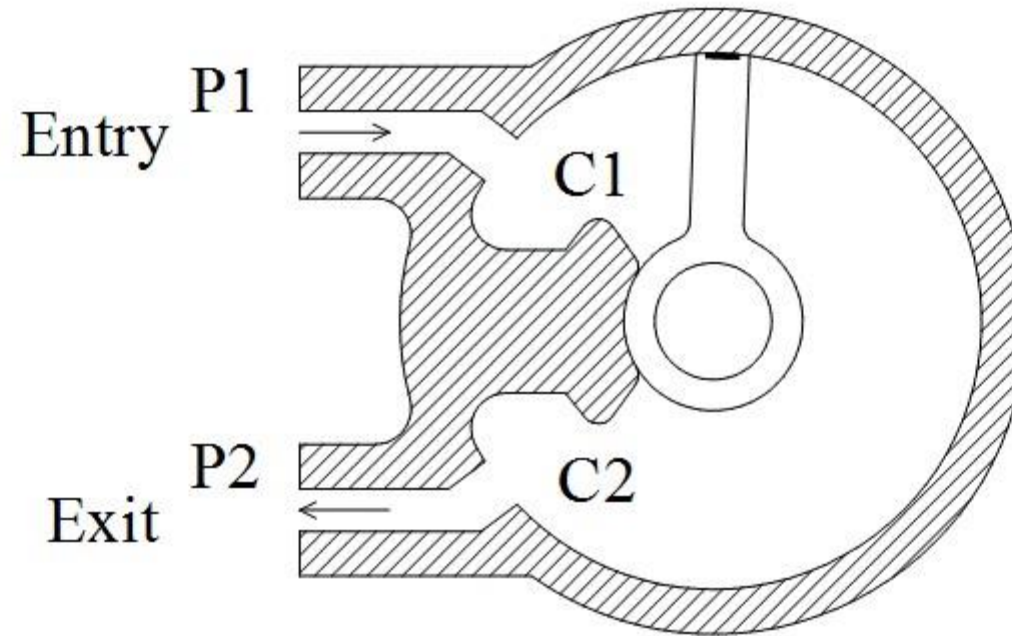
- 1. They produce rotary motion as its output.**
- 2. Types of rotary actuators**
  - a) Rotary vane actuator**
  - b) Rack and pinion actuator**
    - Single cylinder rack and pinion actuator**
    - Double cylinder rack and pinion actuator**

# Rotary Vane Actuator



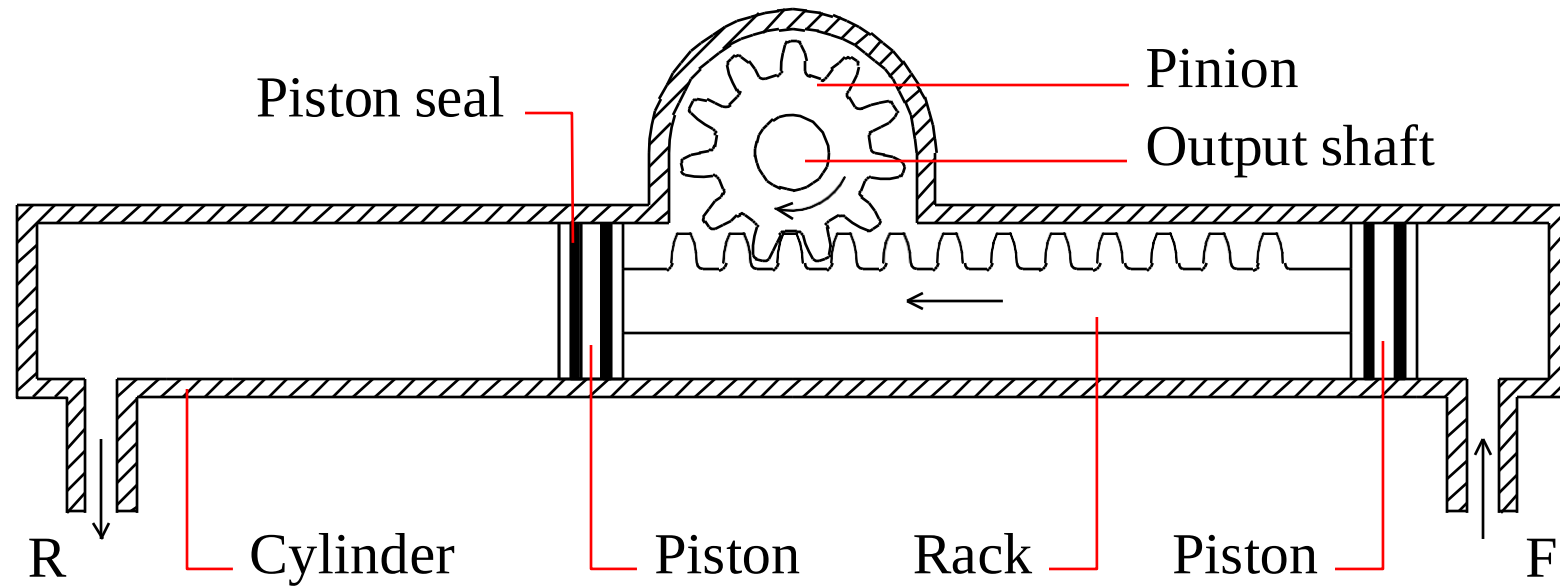
# Rotary Vane Actuator

Clockwise Rotation



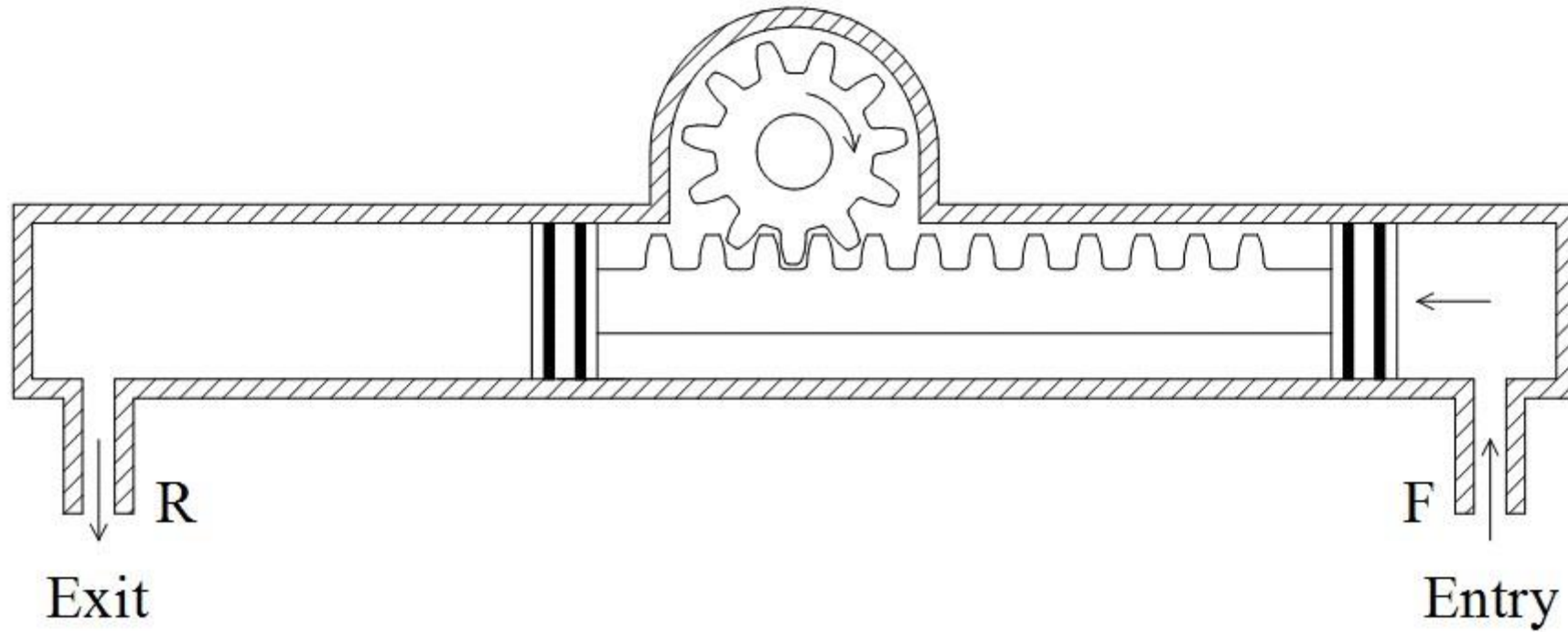
Animation

# Single Cylinder Rack and Pinion Actuator



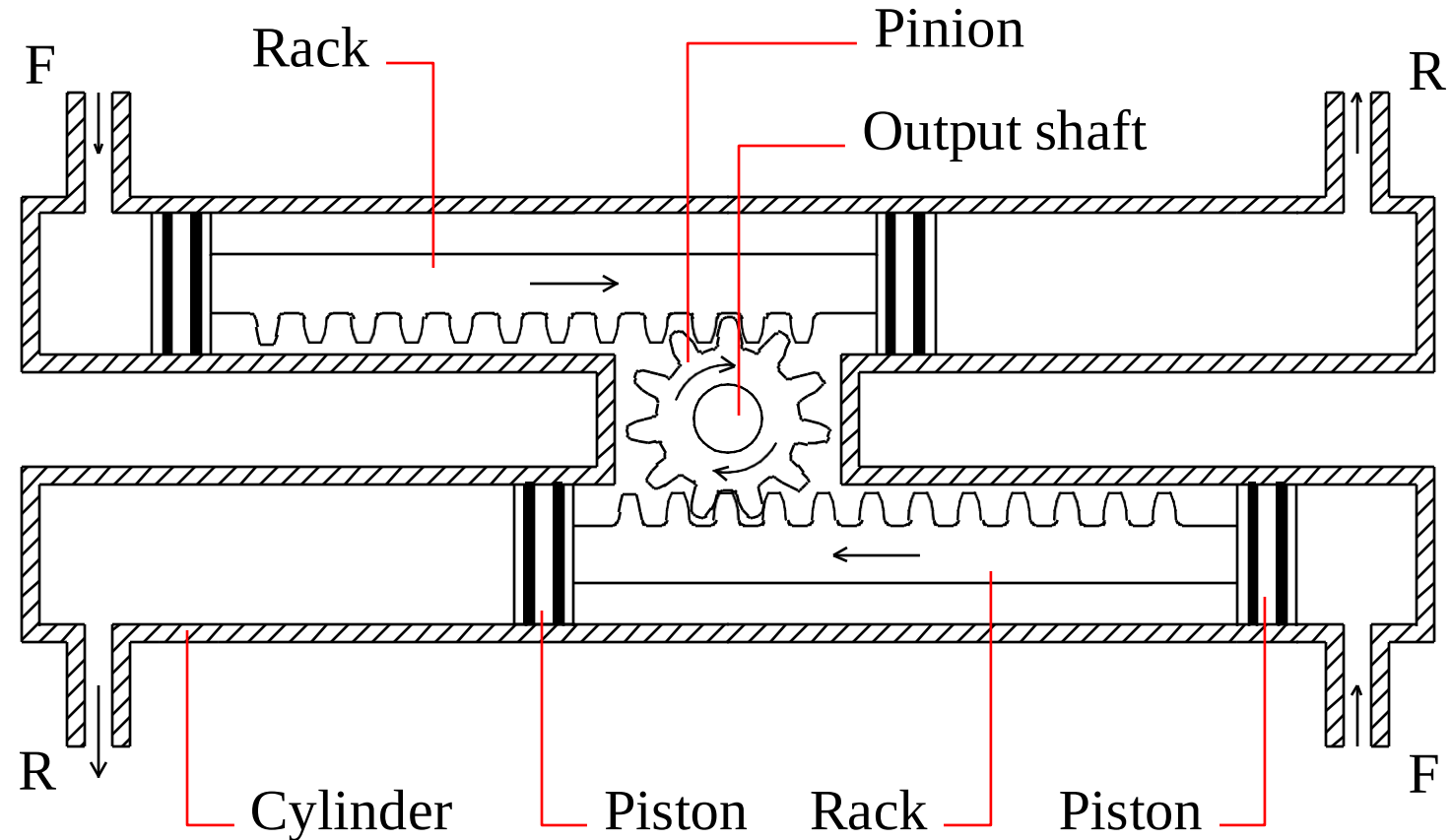
# Single Cylinder Rack and Pinion Actuator

Clockwise Rotation

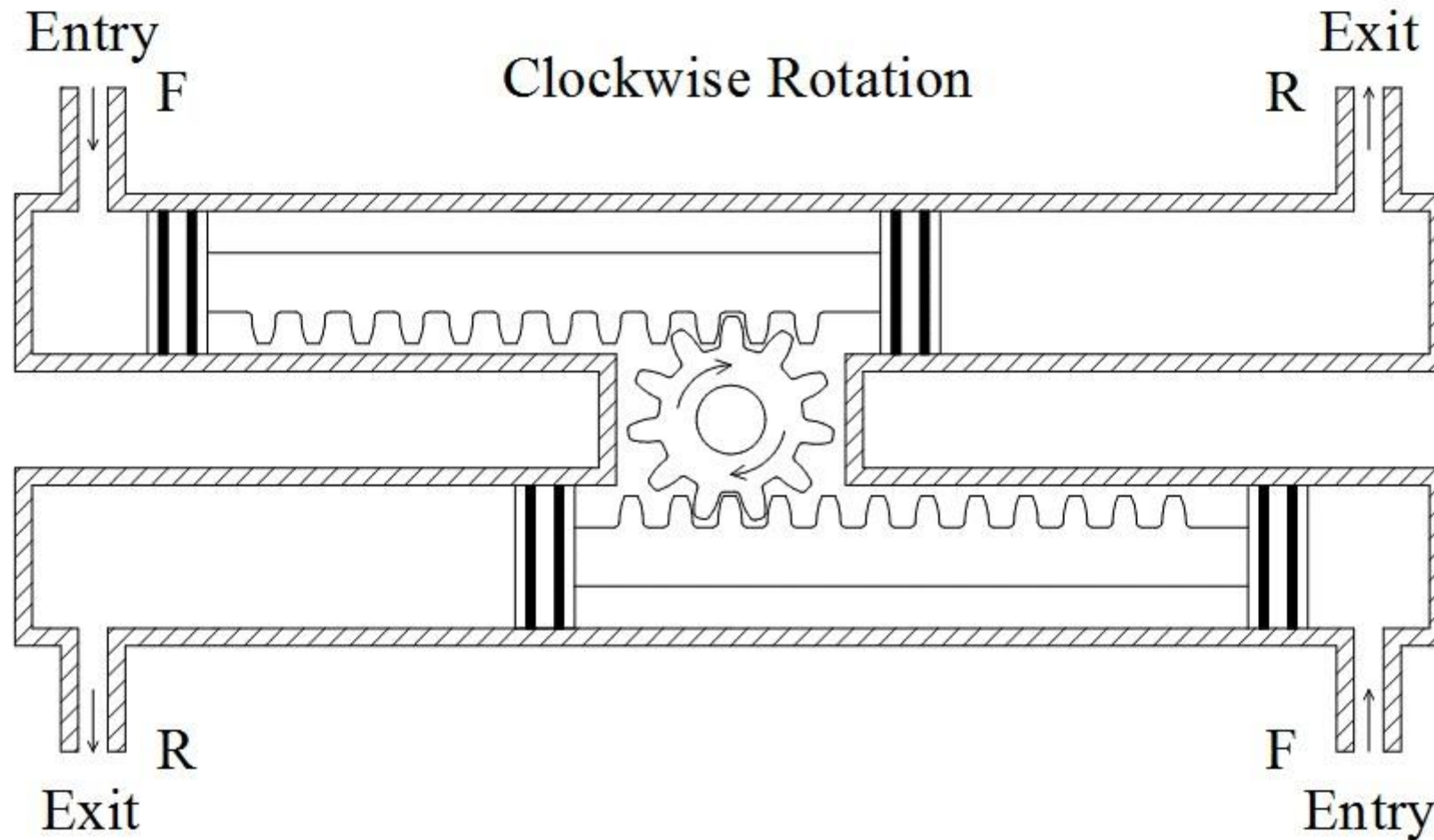


Animation

# Double Cylinder Rack and Pinion Actuator

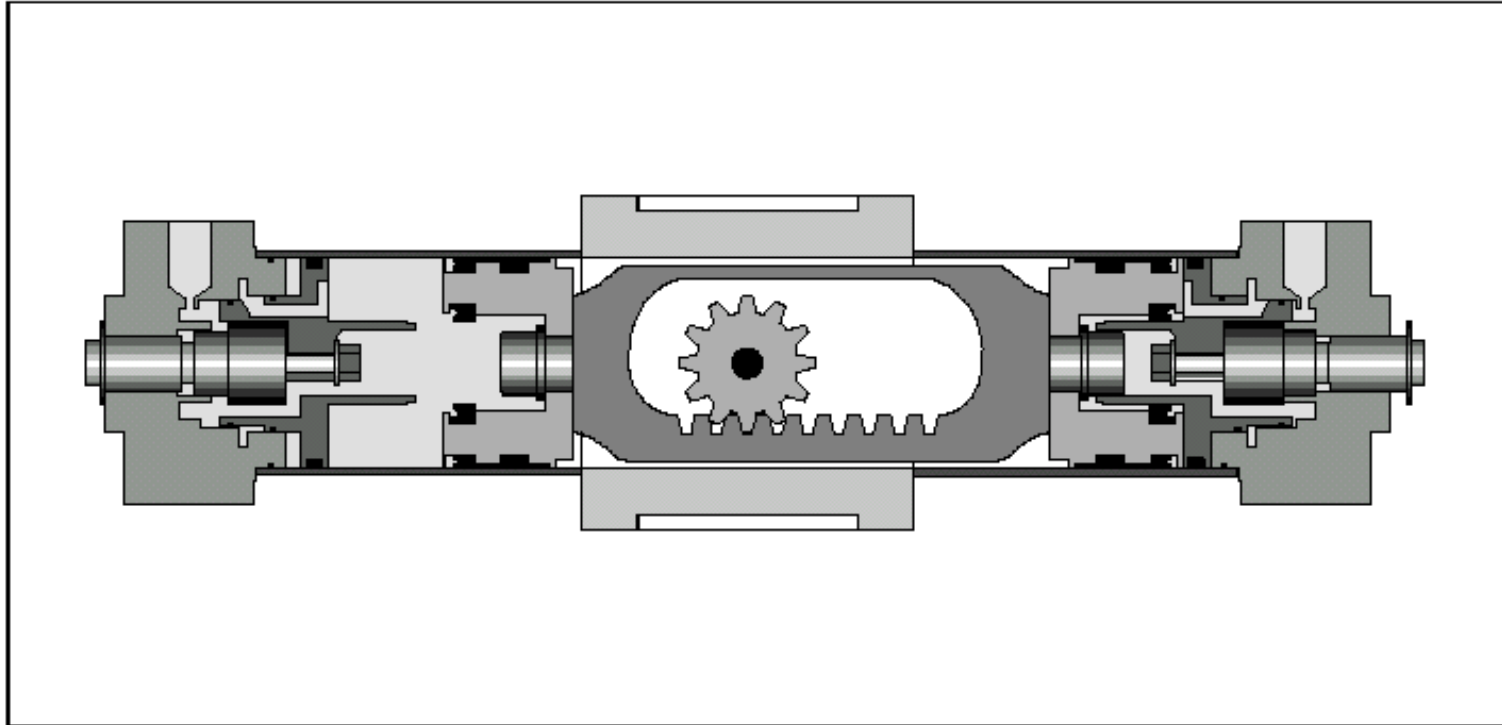


# Double Cylinder Rack and Pinion Actuator

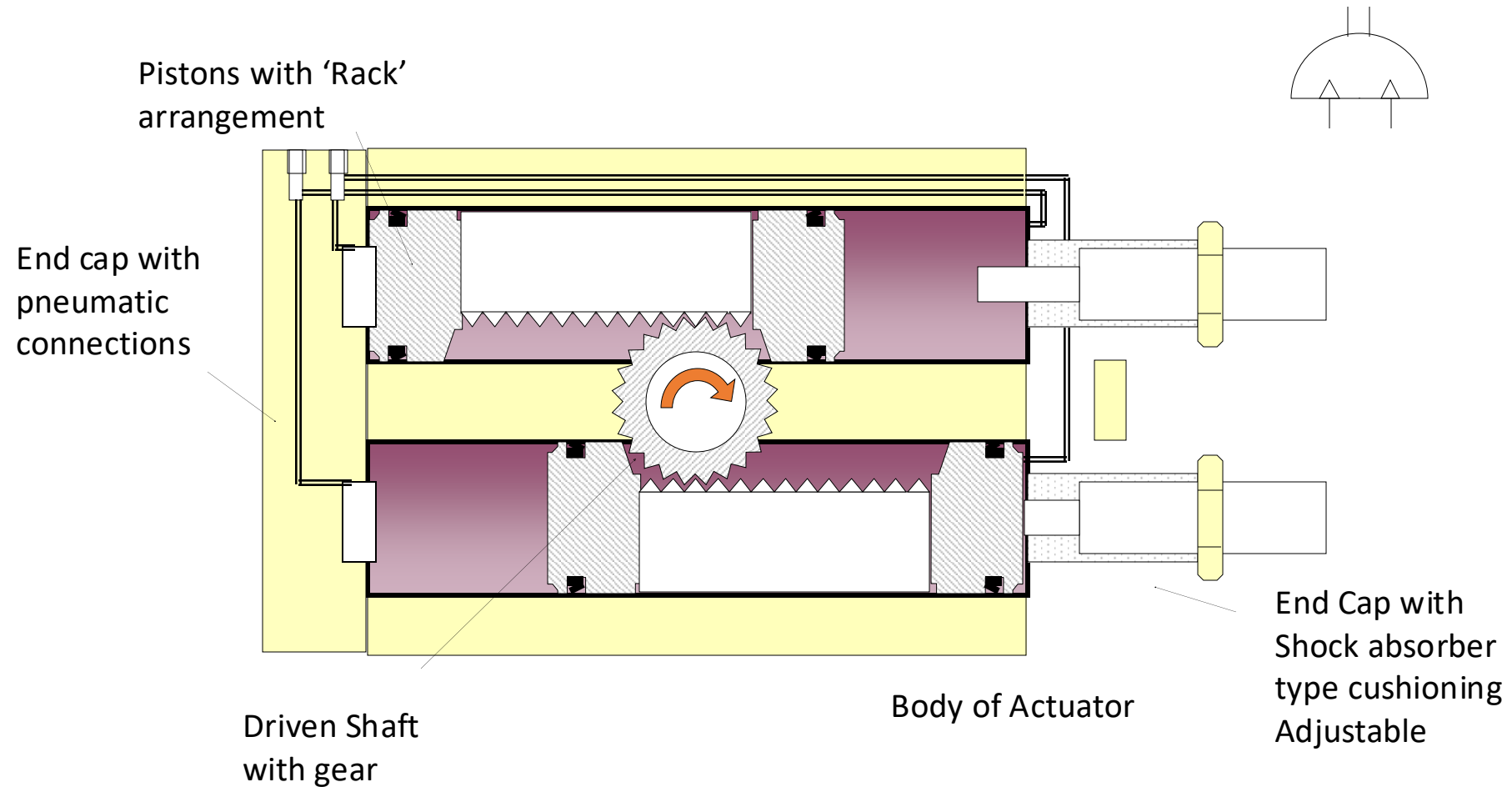


Animation

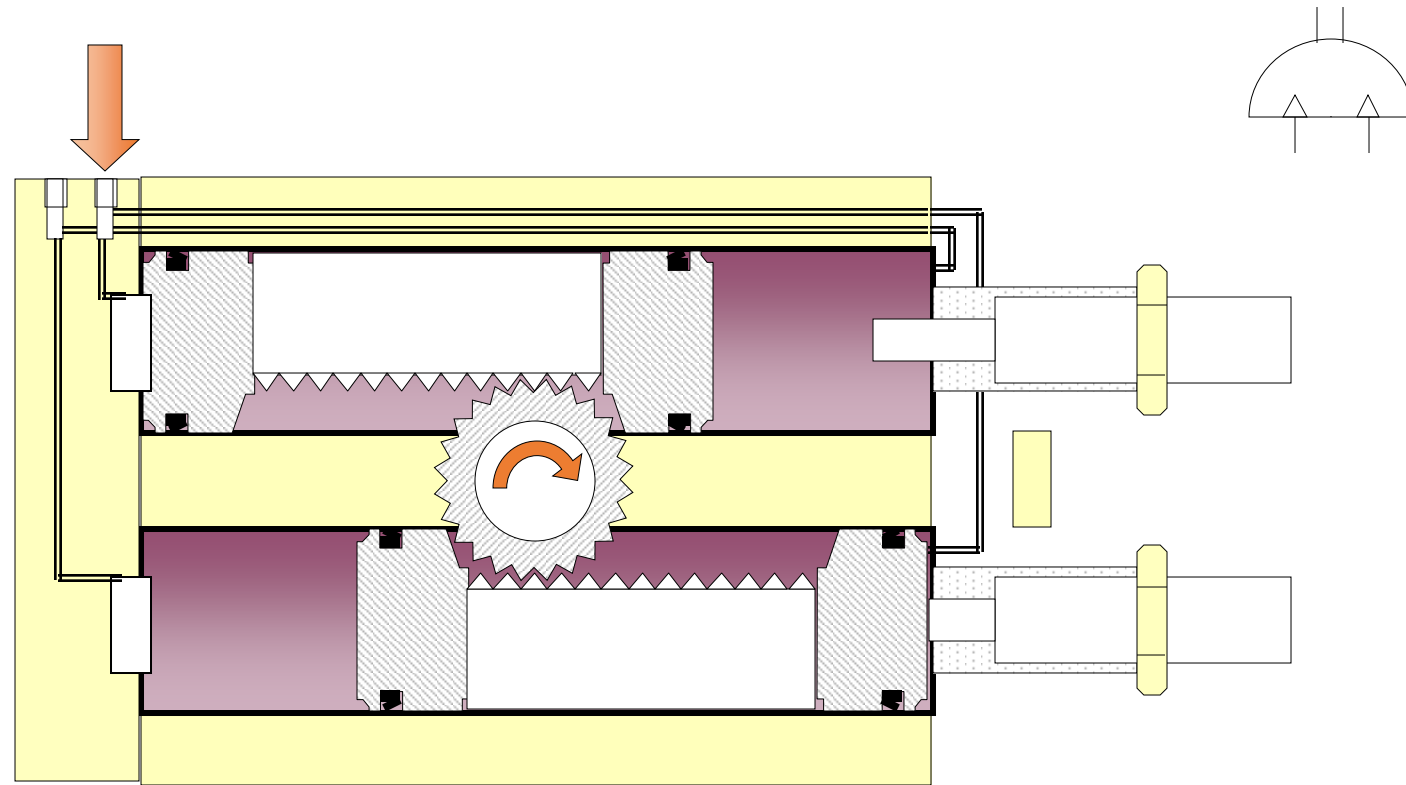
# Rack and pinion rotary actuator



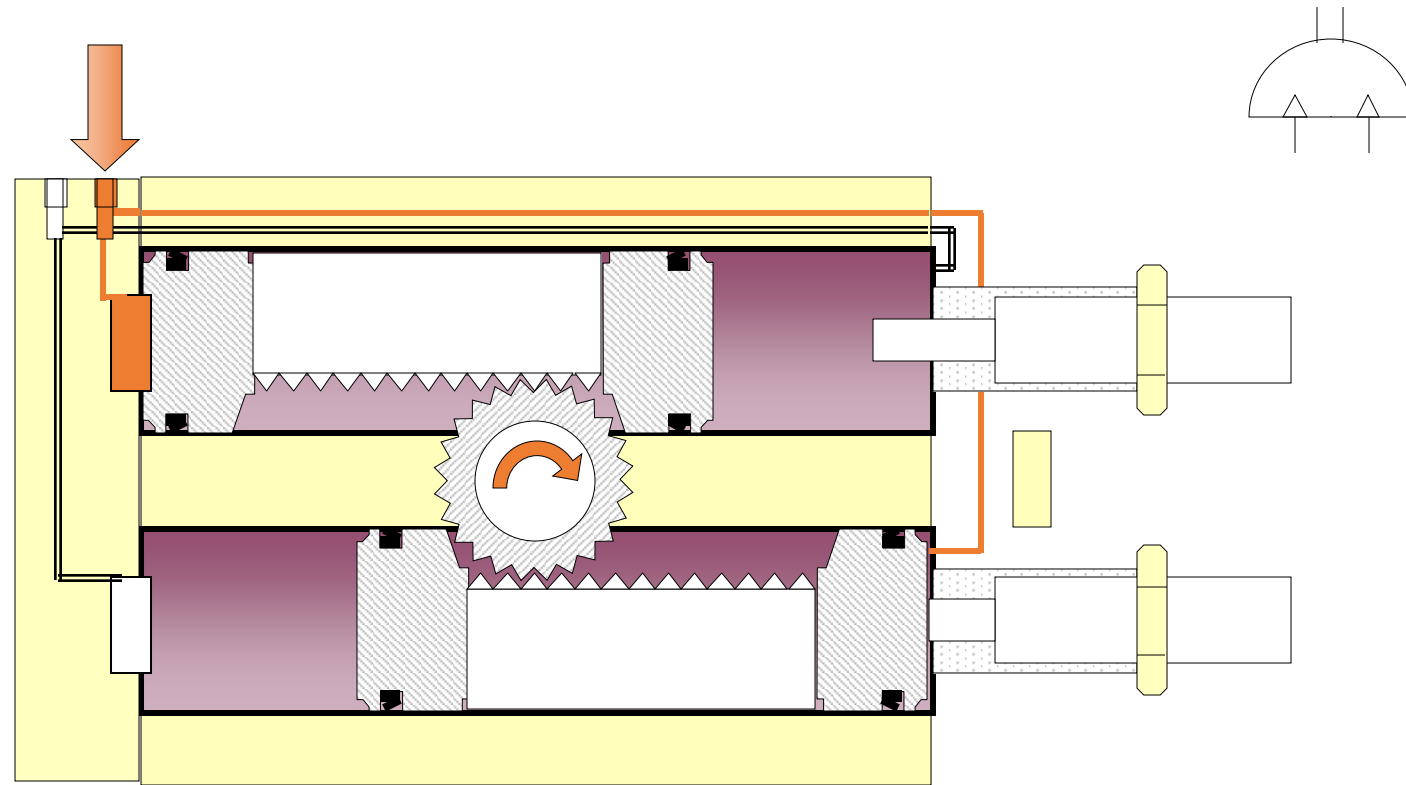
# Rack and pinion rotary actuator



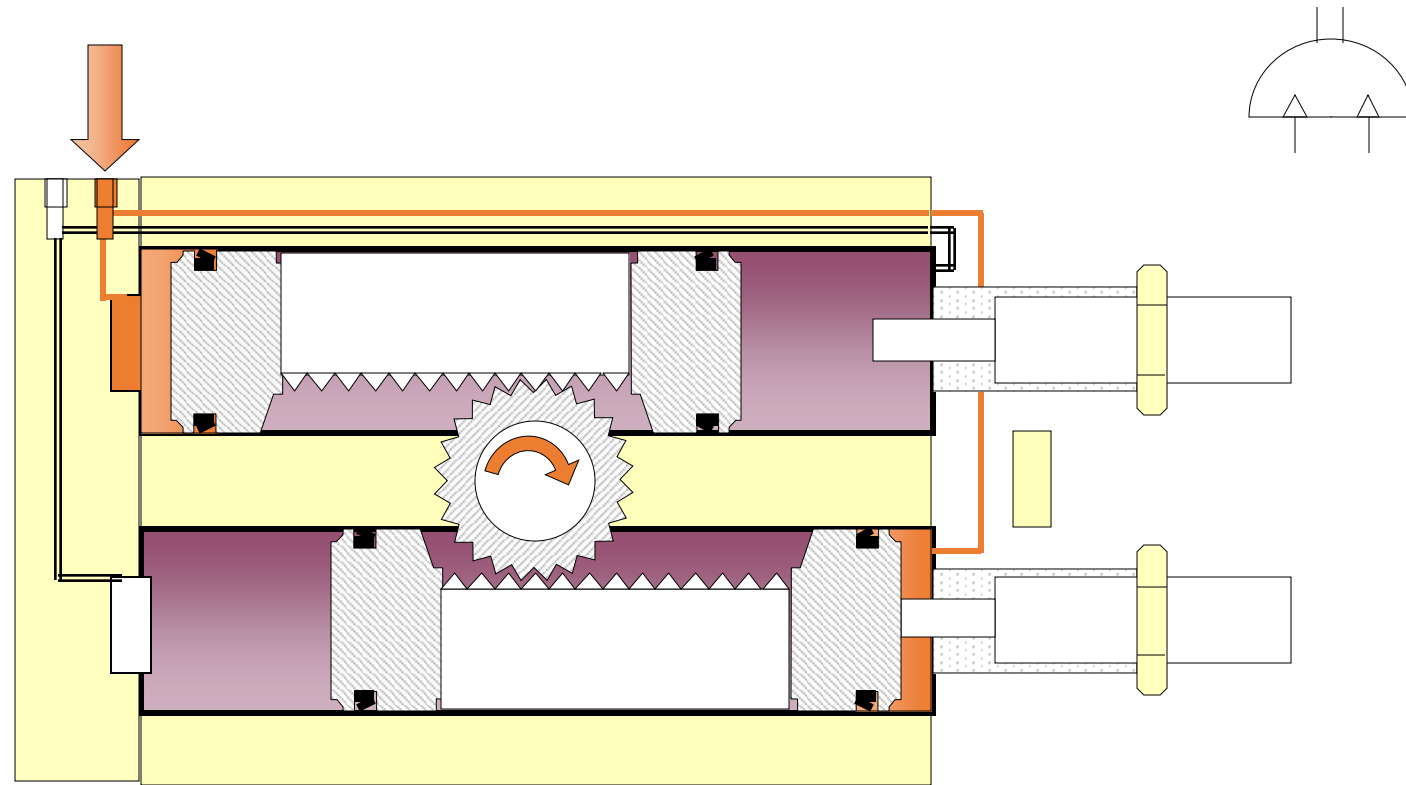
# Rack and pinion rotary actuator



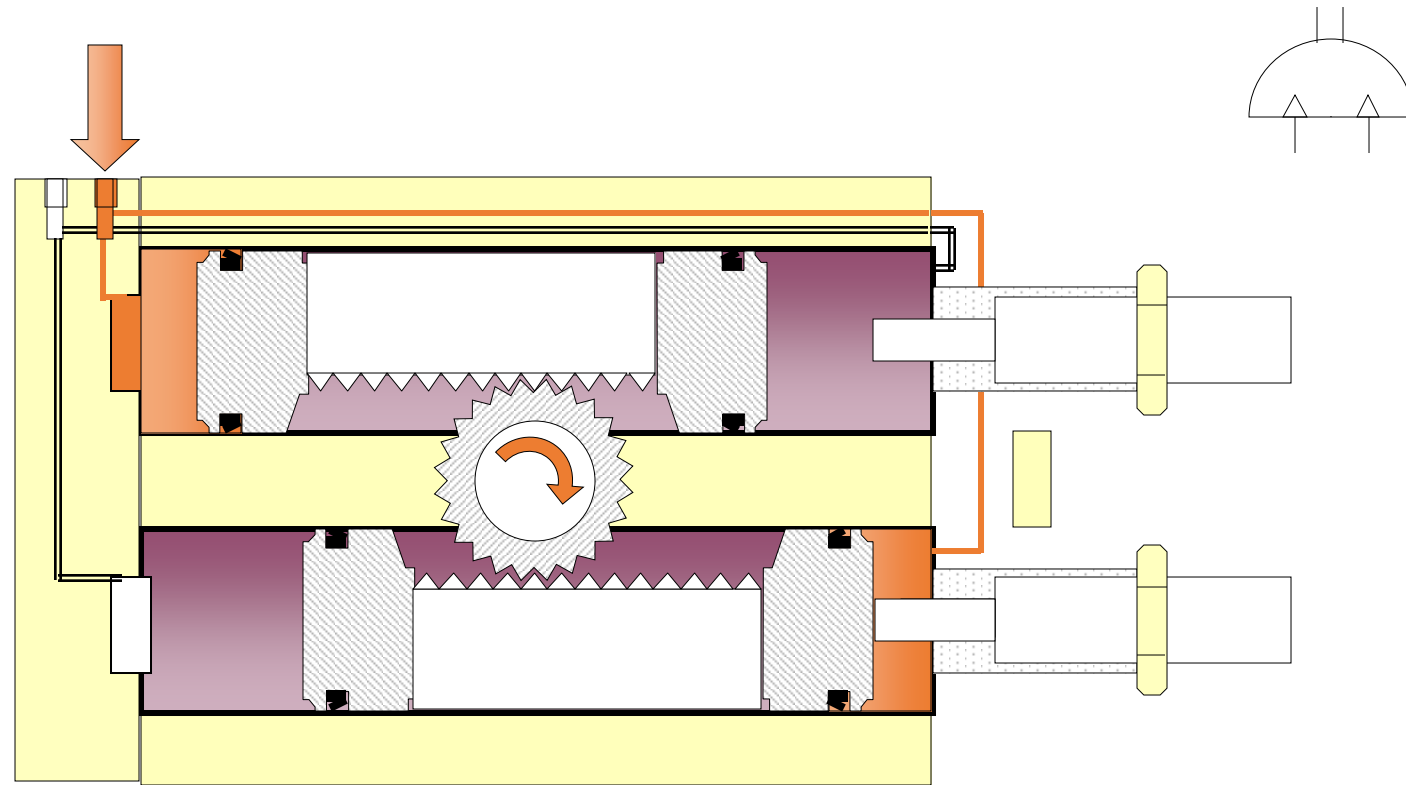
# Rack and pinion rotary actuator



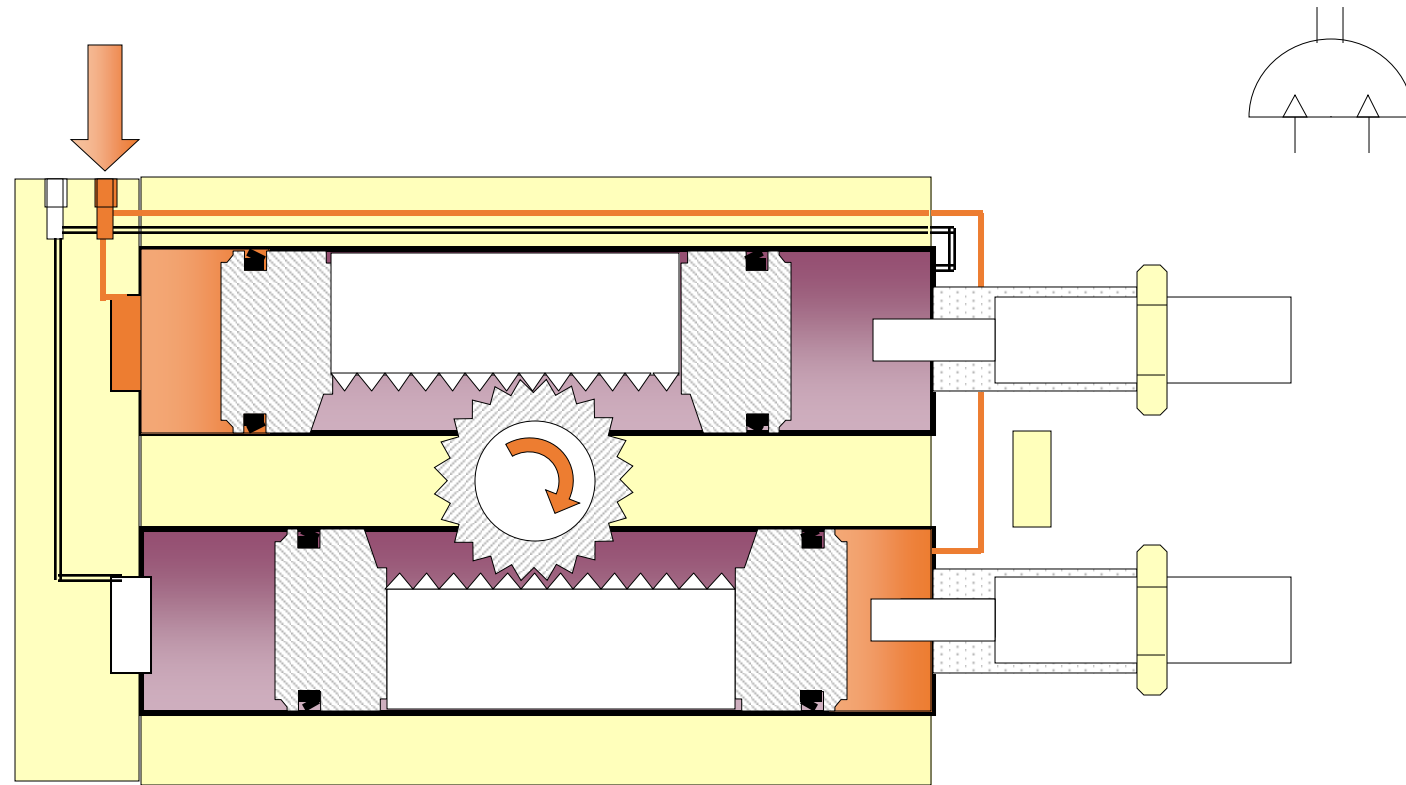
# Rack and pinion rotary actuator



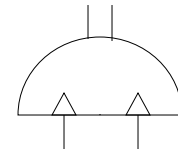
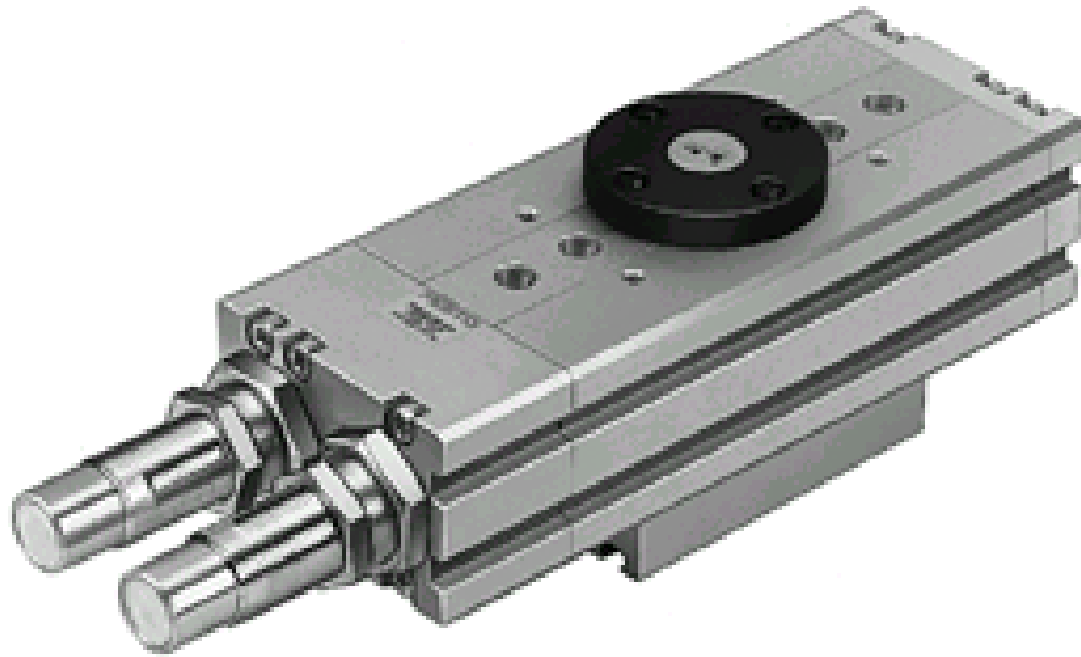
# Rack and pinion rotary actuator



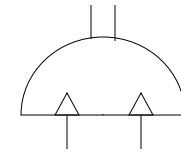
# Rack and pinion rotary actuator



# Rack and pinion rotary actuator



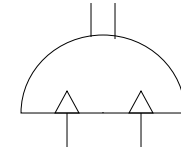
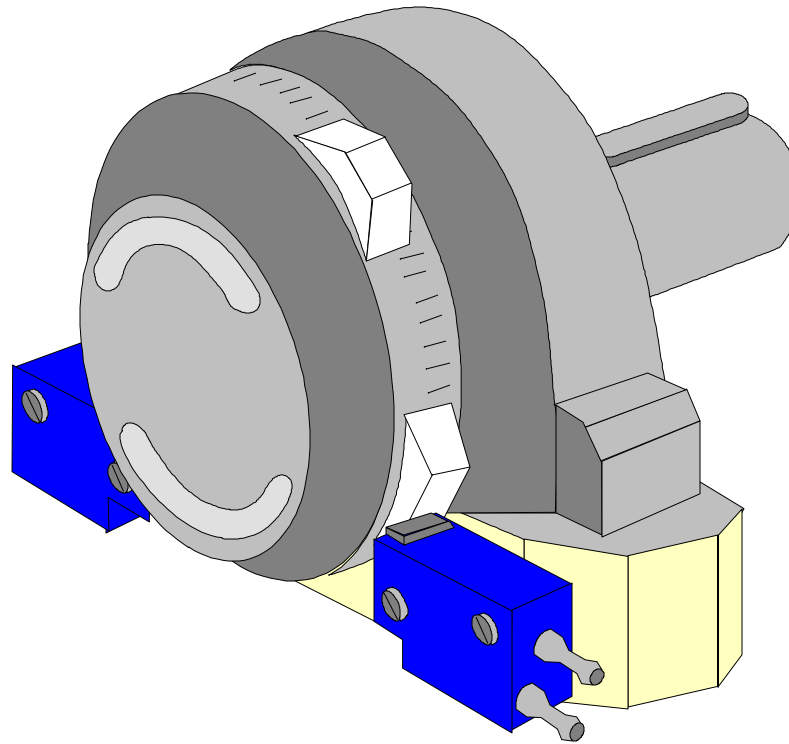
# Rack and pinion rotary actuator



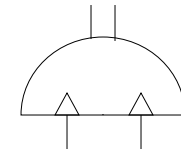
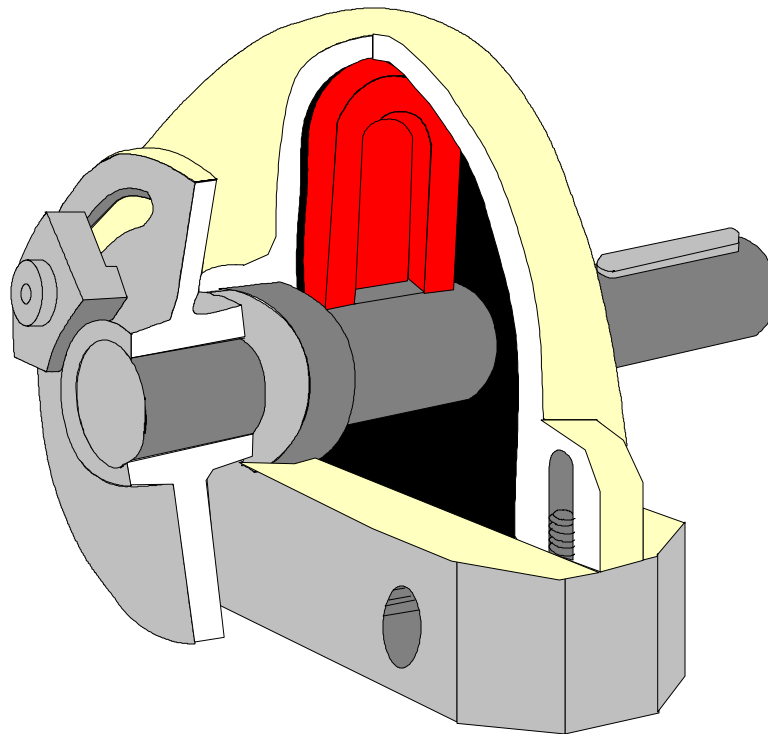
# Vane type rotary actuator



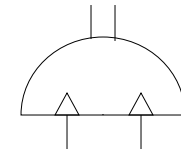
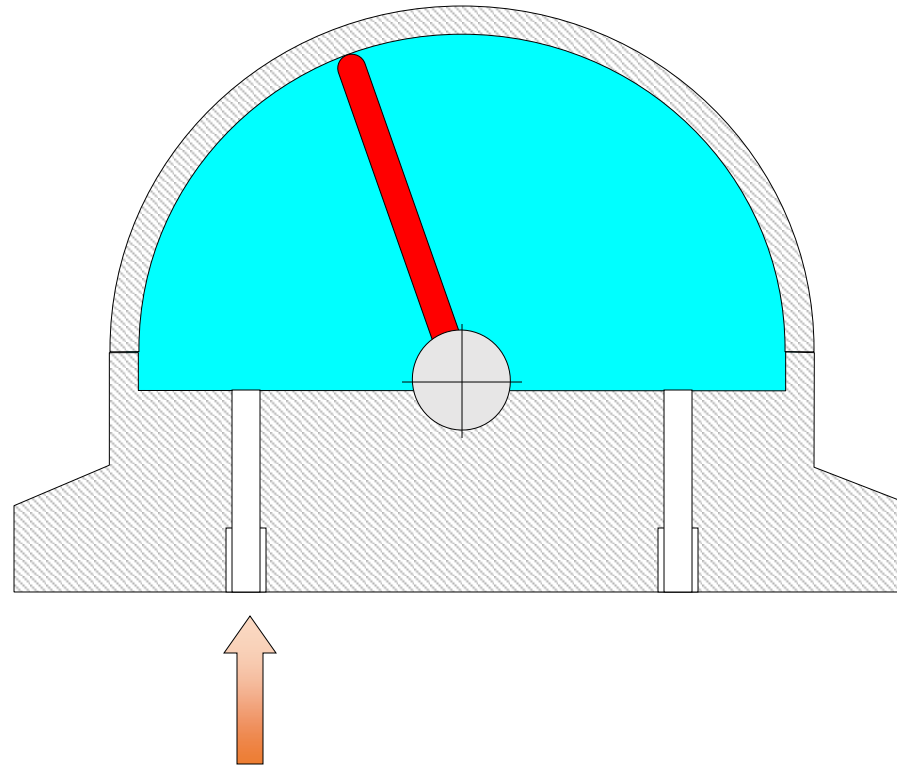
# Vane type rotary actuator



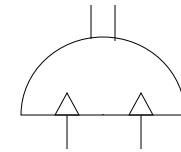
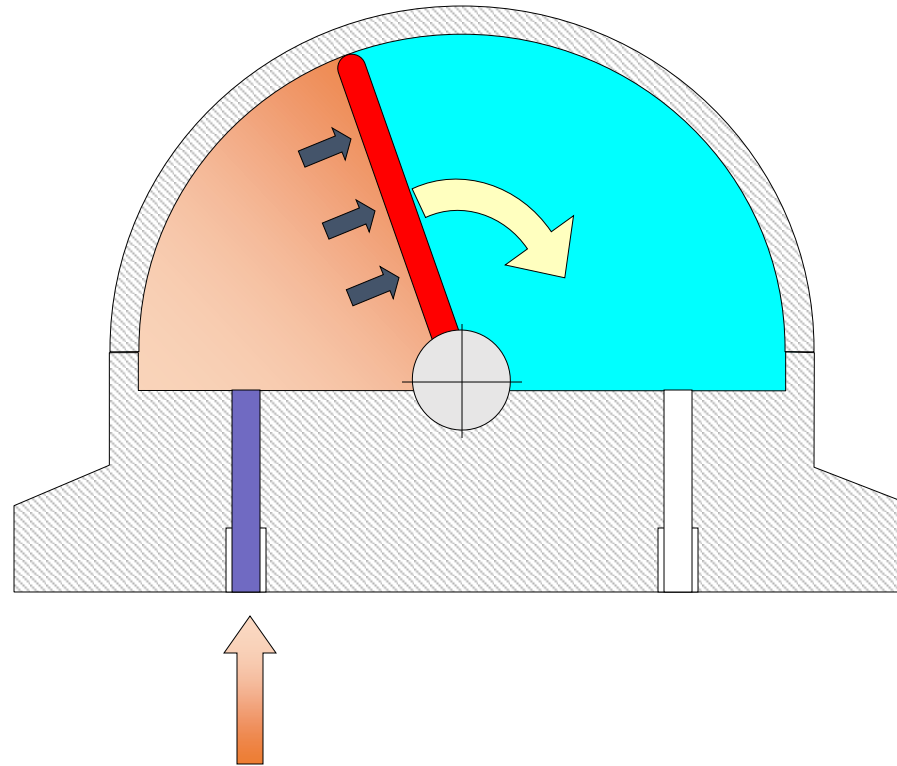
# Vane type rotary actuator



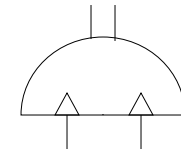
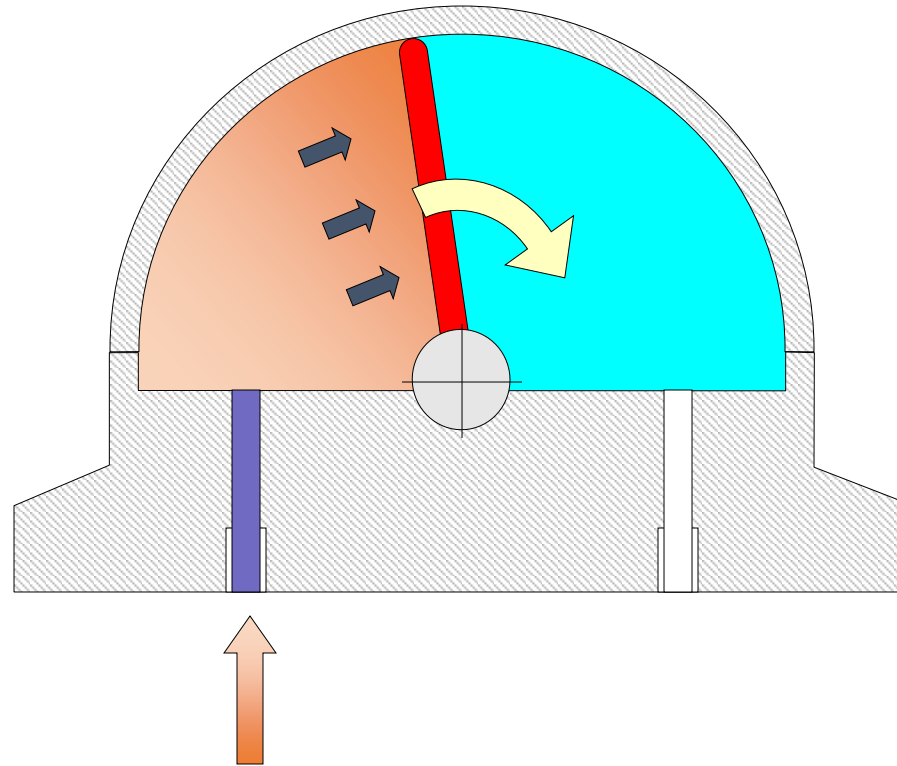
# Vane type rotary actuator



# Vane type rotary actuator



# Vane type rotary actuator



# Linear and rotary actuator combination

